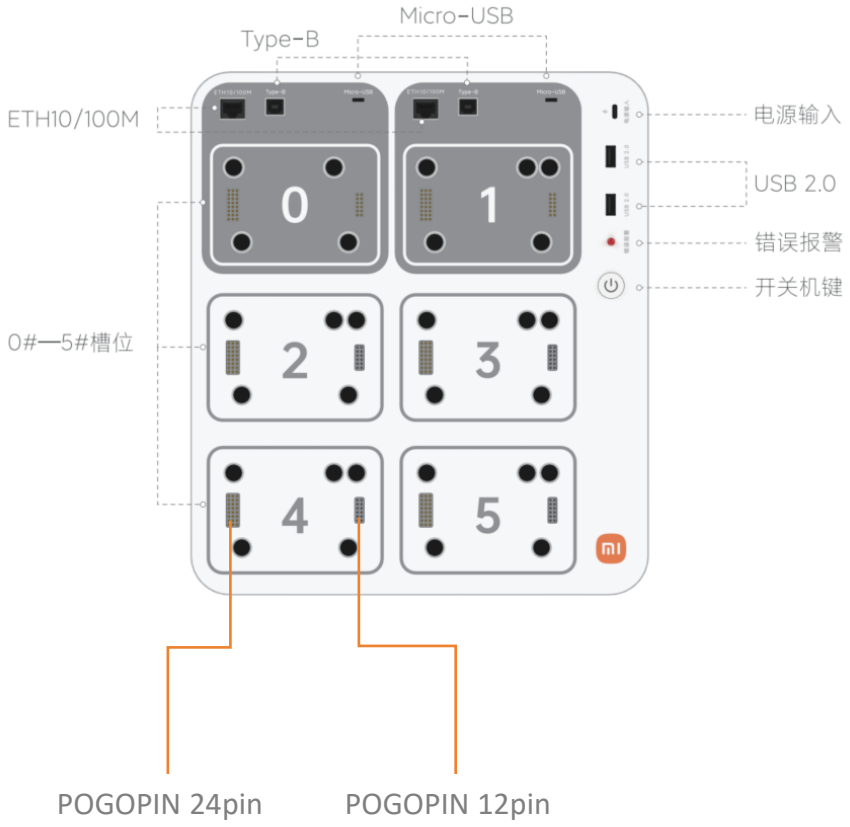
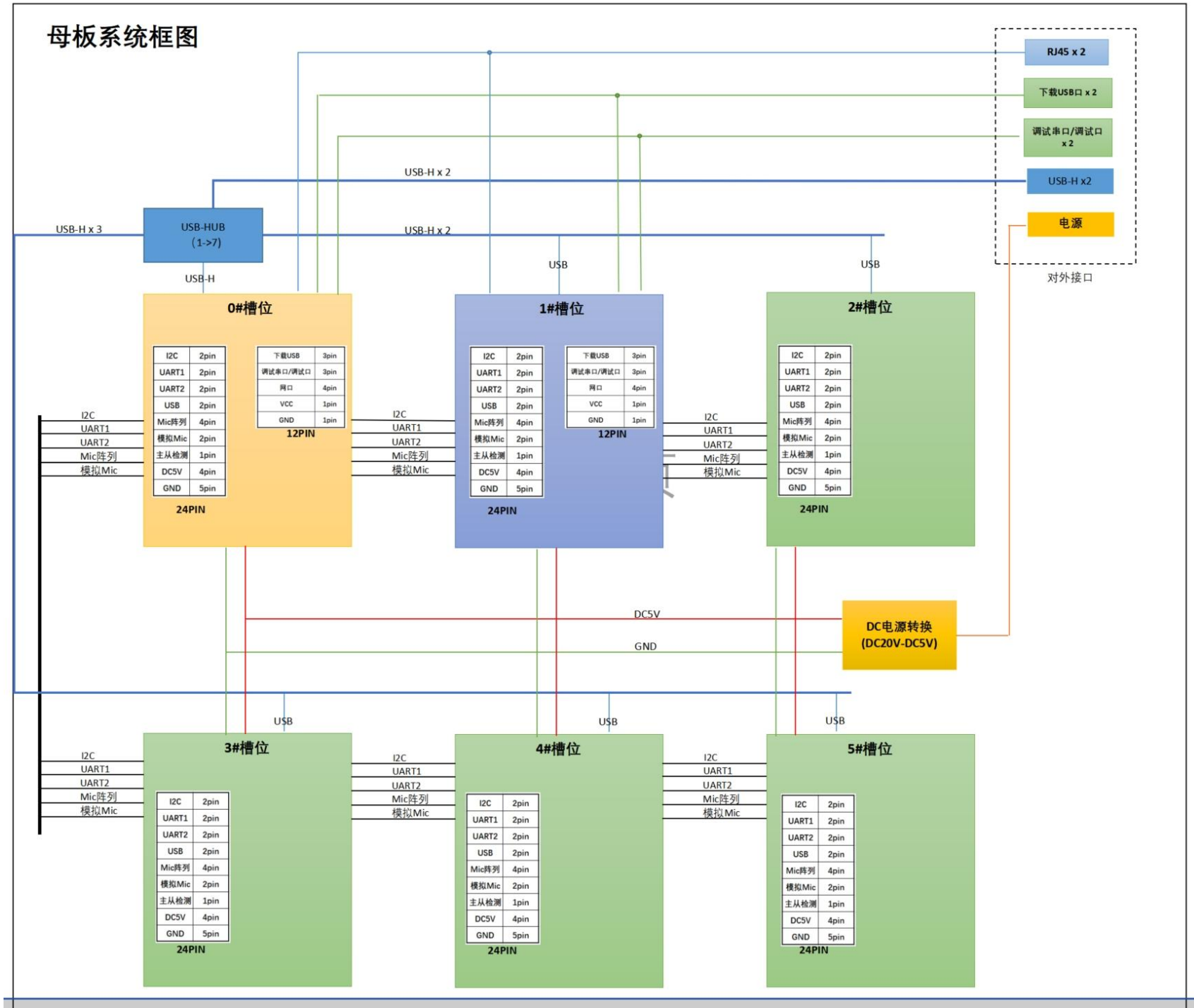
M1母板接口


Micro-USB
Type-B
ETH10/100M
电源输入
USB 2.0
错误报警
开关机键
0#—5#槽位
0
1
2
3
4
5
POGOPIN 24pin
POGOPIN 12pin


母板系统框图
RJ45 x 2
下载USB口 x 2
调试串口/调试口 x 2
USB-H x2
电源
对外接口
USB-H x 2
USB-H x 3
USB-HUB (1->7)
USB-H x 2
USB-H
USB
USB
0#槽位
1#槽位
2#槽位
I2C 2pin
UART1 2pin
UART2 2pin
USB 2pin
Mic阵列 4pin
模拟Mic 2pin
主从检测 1pin
DC5V 4pin
GND 5pin
24PIN
下载USB 3pin
调试串口/调试口 3pin
网口 4pin
VCC 1pin
GND 1pin
12PIN
I2C
UART1
UART2
Mic阵列
模拟Mic
DC5V
GND
DC电源转换 (DC20V-DC5V)
USB
USB
USB
3#槽位
4#槽位
5#槽位

通用接口说明

POGOPIN（24pin）接口

No.	接口	针脚数	走线	功能	备注
1	I2C	2	引至各槽位	具有I2C功能子板通信用	
2	UART1	2	引至各槽位	接C1通信子板用	受UART口的电气特性限制，每种组合中只能有1个C1子板
3	UART2	2	引至各槽位	接C2通信子板用	受UART口的电气特性限制，每种组合中只能有1个C2子板
4	USB	2	引至母板	对外引出USB主设备	0#槽位的USB是主设备，其余槽位的USB设备都是从设备
5	Mic阵列	4	引至各槽位	PDM接口，接4路mic阵列用	
6	模拟Mic	2	引至各槽位	模拟接口，接1路模拟mic用	
7	主从检测	1	引至母板	U板、C板通过该脚位电平确认主从	
8	DC5V	4	引至母板	5V电源，每个槽位最大提供3A电流	
9	GND	5	引至母板	电源GND	考虑到端子有多种数字信号接口，为数字信号就近提供0V电位和子板供电电流回路需求
合计pin数		24			

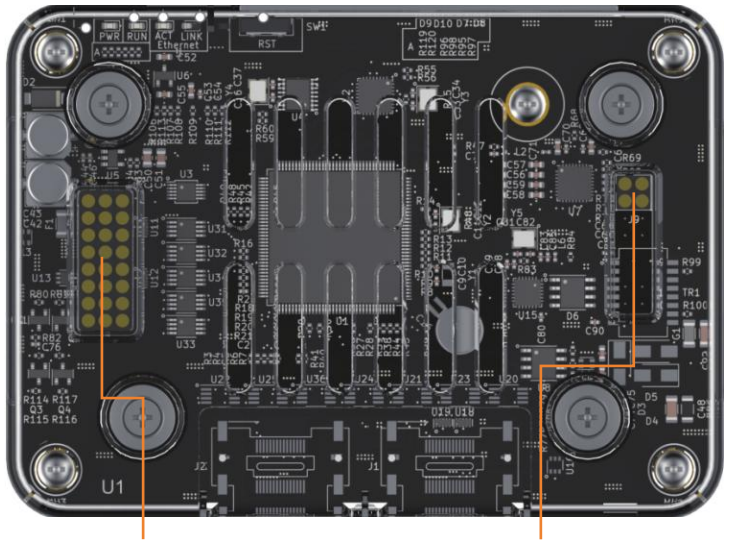
POGOPIN（12pin）接口

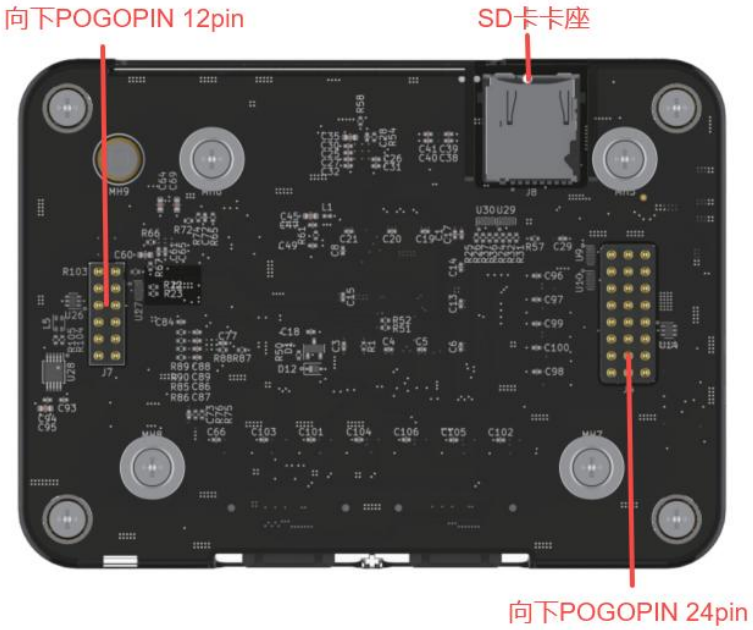
No.	接口	针脚数	走线	功能	备注
1	下载USB	3	引至母板	U2、U3程序下载用(D+ D- VUSB)	
2	调试串口/调试口	3	引至母板	U1、C1、C2、C3程序下载调试用	
3	网口	4	引至母板	U板对外引出的网口	
4	VCC	1	引至母板	调试器电源+	
5	GND	1	引至母板	调试器电源-	
合计pin数		12			

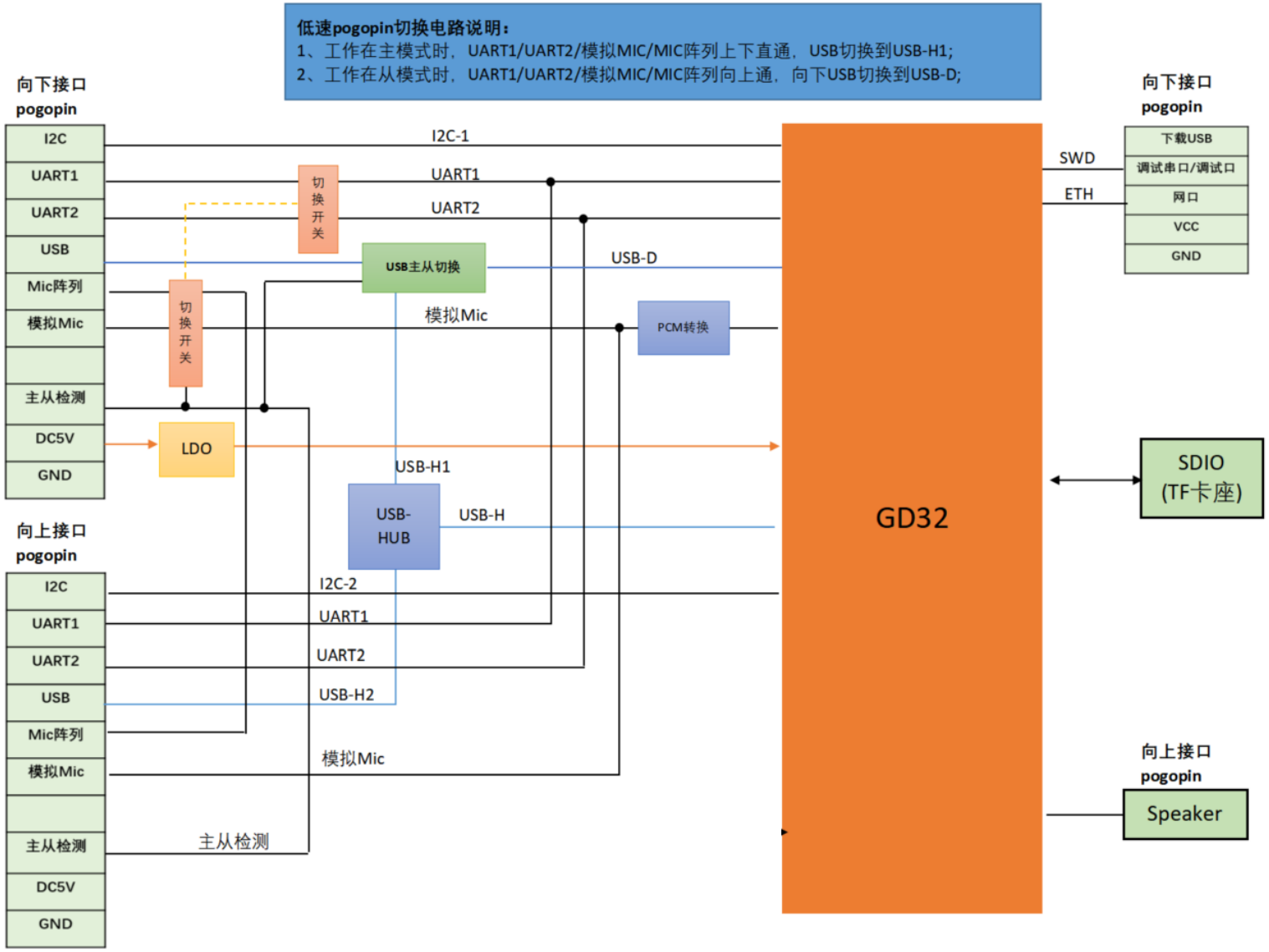
FPC高速接口

No.	接口	针脚数	功能	备注
1	SPI	4	具有SPI功能子板通信用	
2	RS485	2	具有RS485功能子板通信用	
3	UART3	2	新增串口	
4	UART4	2	新增串口	
7	GPIO	5	通用输入输出口	
8	GND	8	电源及信号地	考虑到端子有多种数字信号接口，为数字信号就近提供0V电位和子板供电电流回路需求
9	DC5V	7	5V电源，最大提供2.5A电流	
合计pin数		30		

U1接口


U1
向上POGOPIN 24pin
向上POGOPIN 4pin

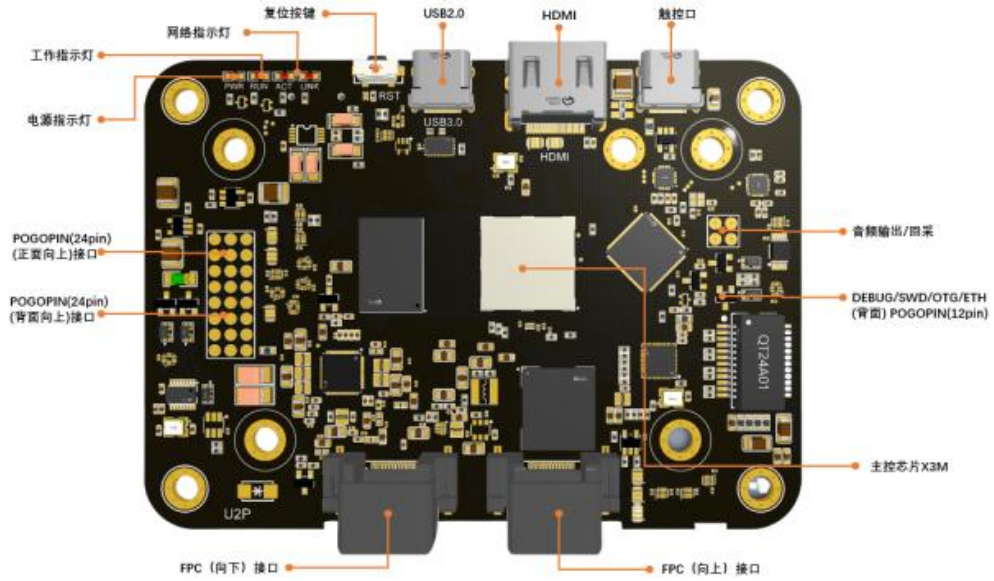

向下POGOPIN 12pin
SD卡卡座
向下POGOPIN 24pin

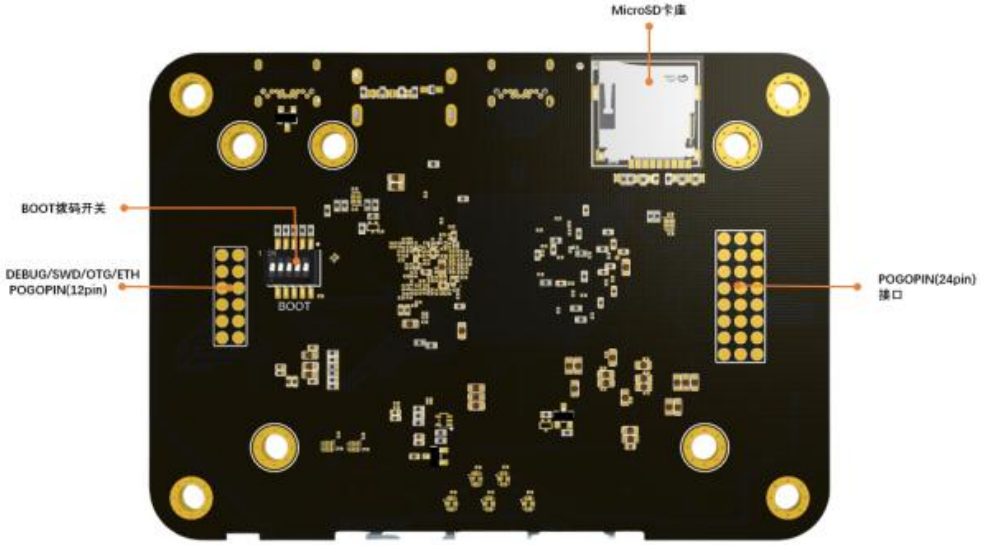

低速pogopin切换电路说明：
1、工作在主模式时，UART1/UART2/模拟MIC/MIC阵列上下直通，USB切换到USB-H1；
2、工作在从模式时，UART1/UART2/模拟MIC/MIC阵列向上通，向下USB切换到USB-D；
向下接口
pogopin
I2C
UART1
UART2
USB
Mic阵列
模拟Mic
主从检测
DC5V
GND
I2C-1
UART1
UART2
切换开关
USB主从切换
USB-D
模拟Mic
PCM转换
切换开关
LDO
USB-H1
USB-HUB
USB-H
向上接口
pogopin
I2C
UART1
UART2
USB
Mic阵列
模拟Mic
主从检测
DC5V
GND
I2C-2
UART1
UART2
USB-H2
模拟Mic
主从检测
GD32
SWD
ETH
向下接口
pogopin
下载USB
调试串口/调试口
网口
VCC
GND
SDIO
(TF卡座)
向上接口
pogopin
Speaker

GD32芯片引脚-U1接口对照表

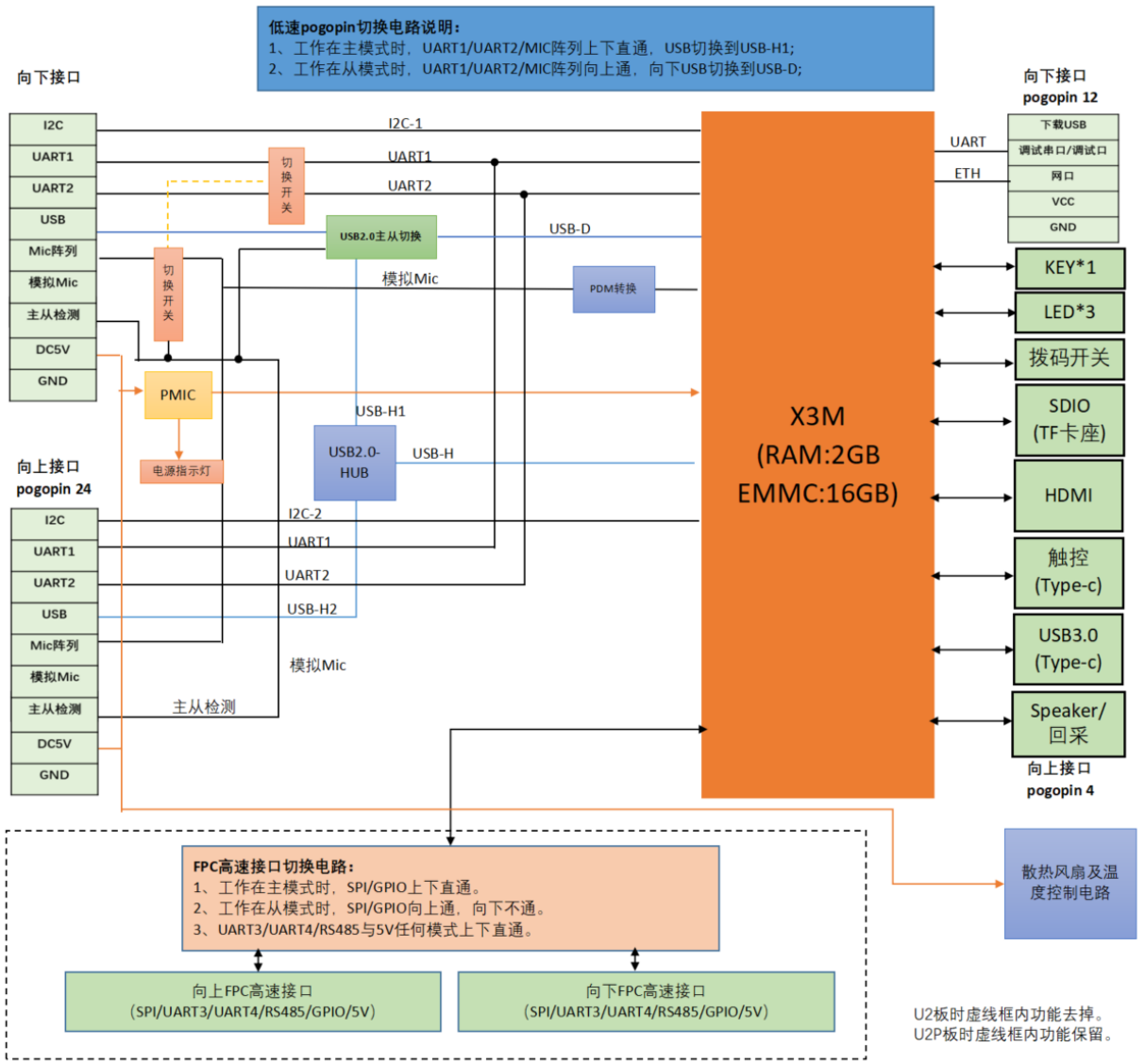
GD32-接口	GD32-IO	U1P-向上POGOPIN（24PIN）接口	U1P-向下POGOPIN（24PIN）接口	U1P-向下POGOPIN（12PIN）接口	U1P-向上POGOPIN（4PIN）接口	备注
I2C0	PB8\PB9	I2C				
I2C1	PF0\PF1		I2C			
USART0	PA9\PA10	UART2	UART2			
USART5	PC6\PC7	UART1	UART1			
USART1	PD5\OD6			DEBUG-USB		
USART2	PD8\PD9\PD12					
UART6	PE7\PE8					
UART7	PE0\PE1					
SPI4	PF6\PF7\PF8\PF9					
ETH	PG11\PG12\PG13\PG14\PC1\PC4\PC5\PA1\PA2\PA7			ETH		
USB	PC0\PC2\PC3\PA3\PA5\PF10\PB0\PB1\PB5\PB10\PB11\PB12\PB13\PA11\PA12	USB	USB			
GPIO	PF11\F12\PF13\PF14\PF15					
SWD	PA13\PA14			SWD		
I2C2	PA8\PB4		MIC\PDM		SPK	在U1P上，I2C2、I2S1接口，通过音频处理芯片，转换为MIC、SPK、PDM三种音频数据接口
I2S1	PD1\PD3\PA6\PB14\PB15					
SDIO	PC8\PC9\PC10\PC11\PC12\PD0\PD2					U1P上的SD卡座接口

U2P接口



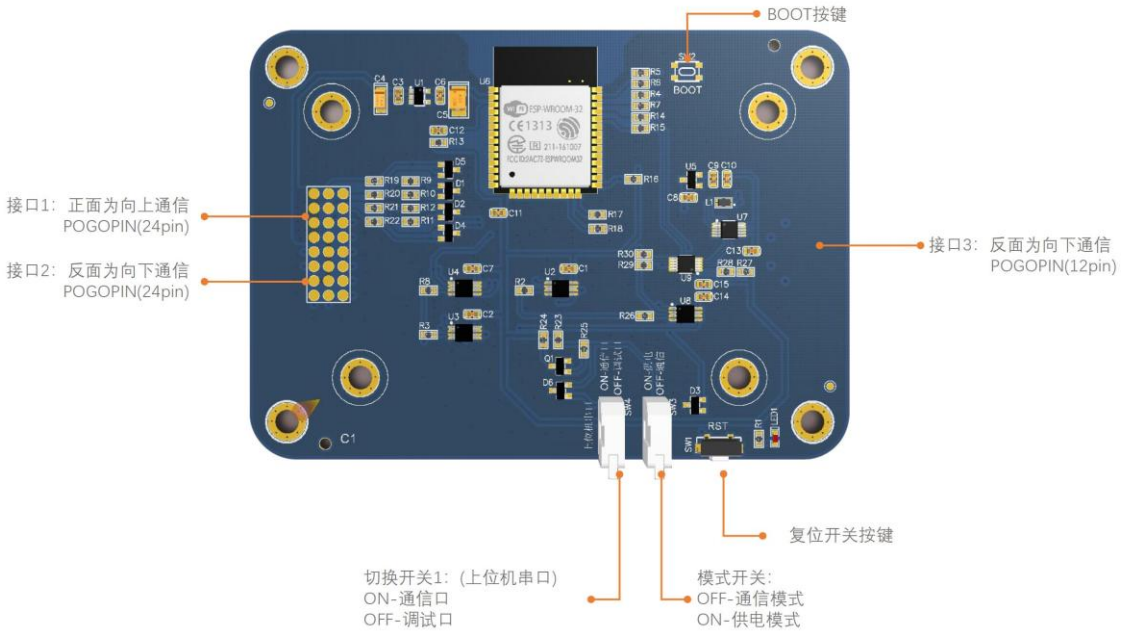


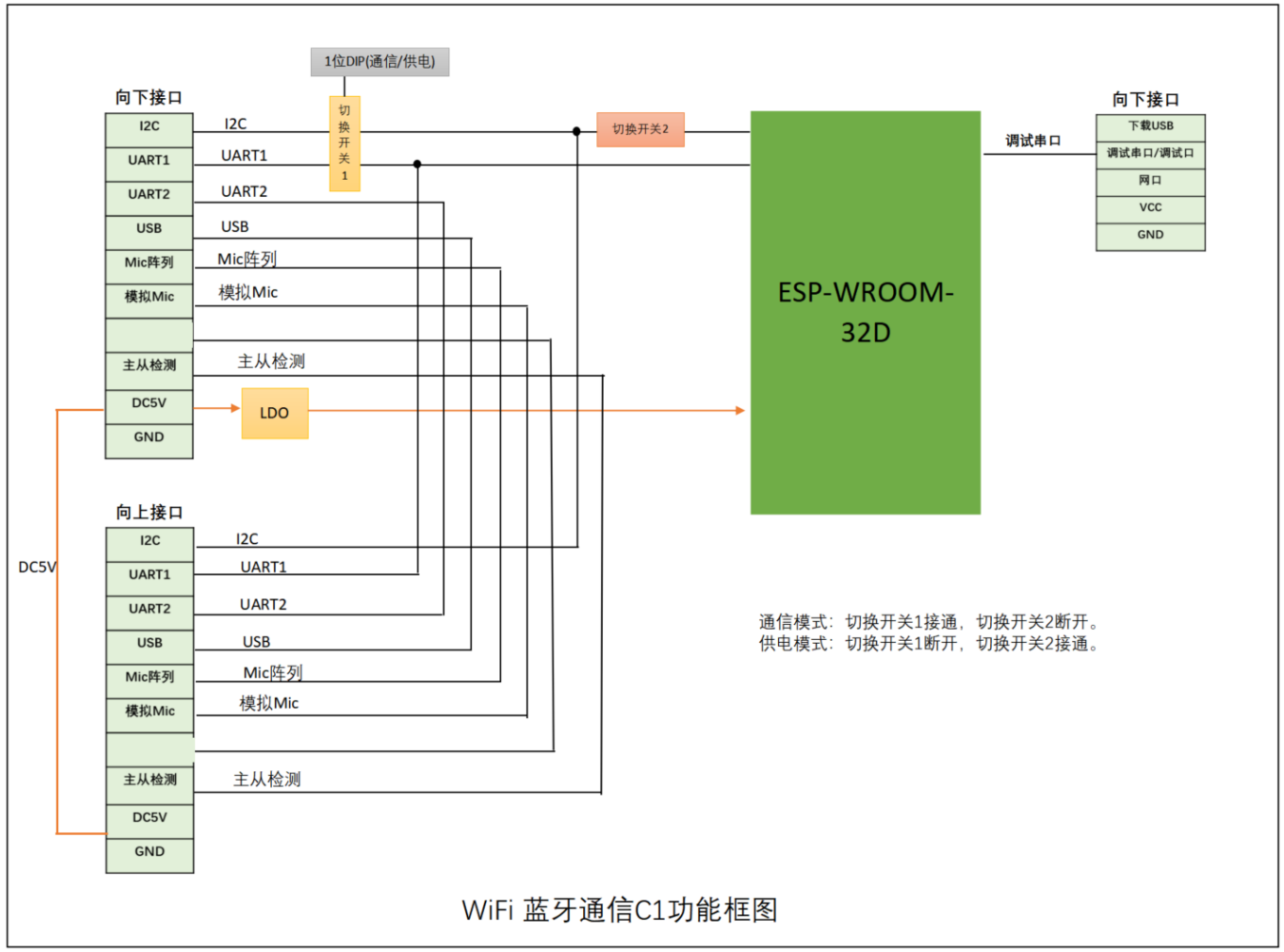
3.1 U2&U2P子板功能框图



计算与存储U2&U2P板功能框图

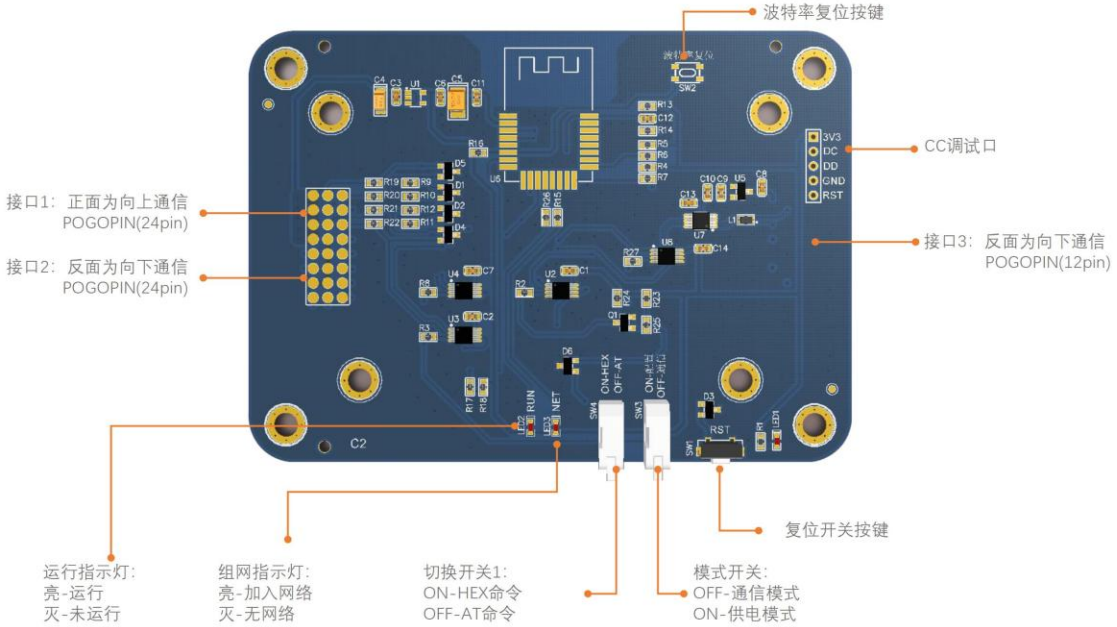
C1接口

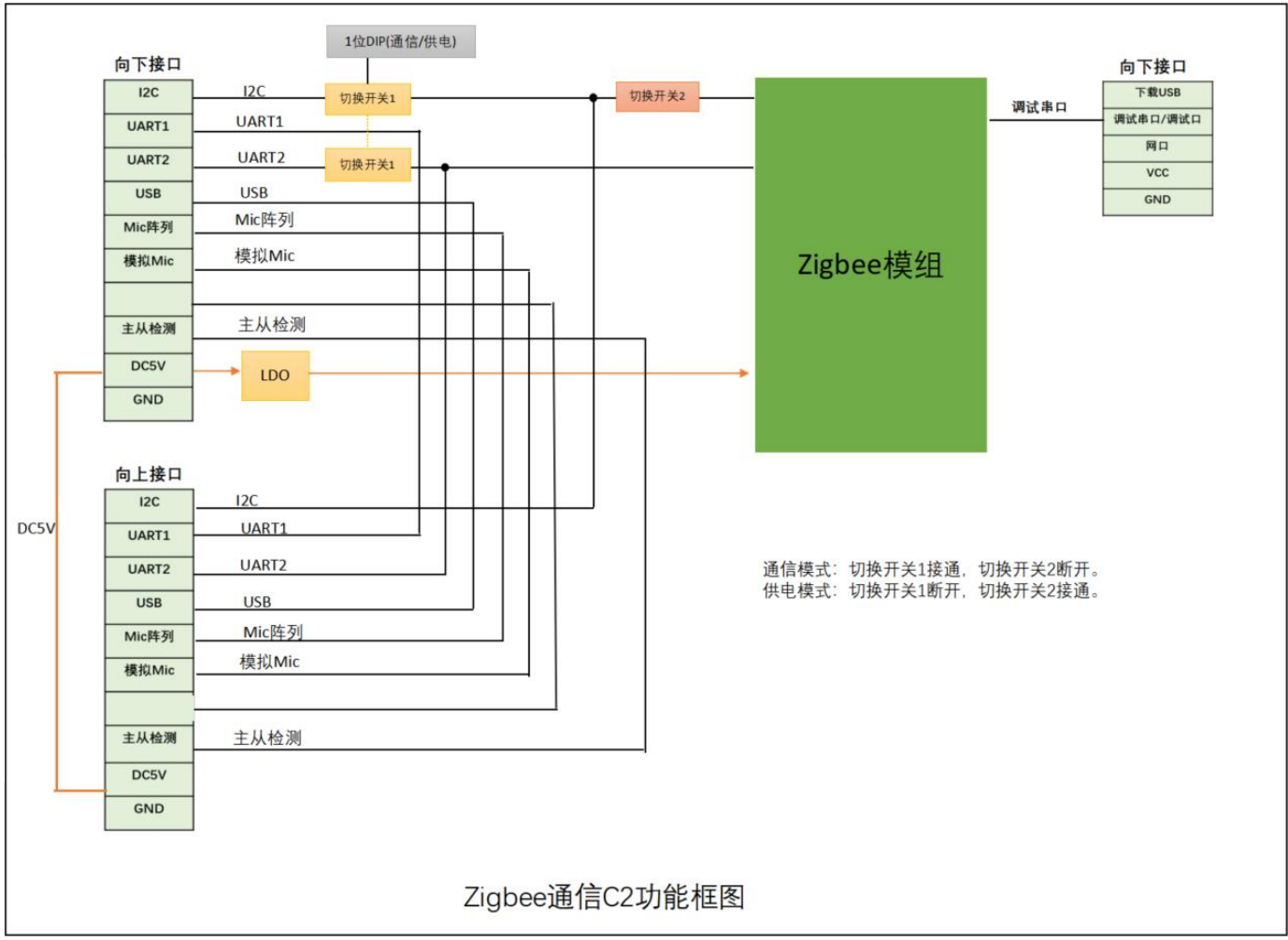




WiFi 蓝牙通信C1功能框图

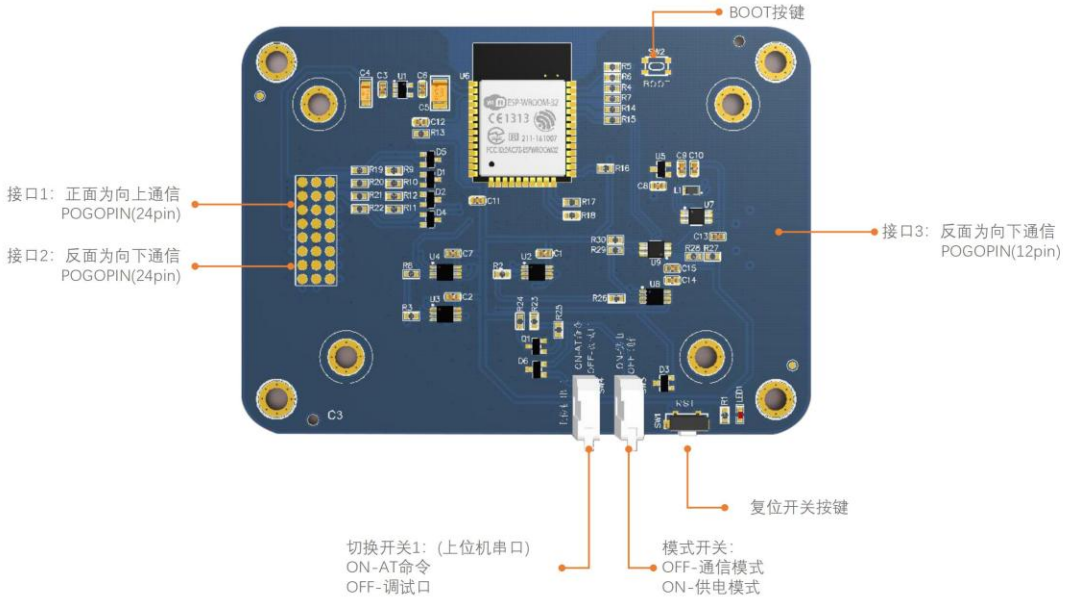
C2接口

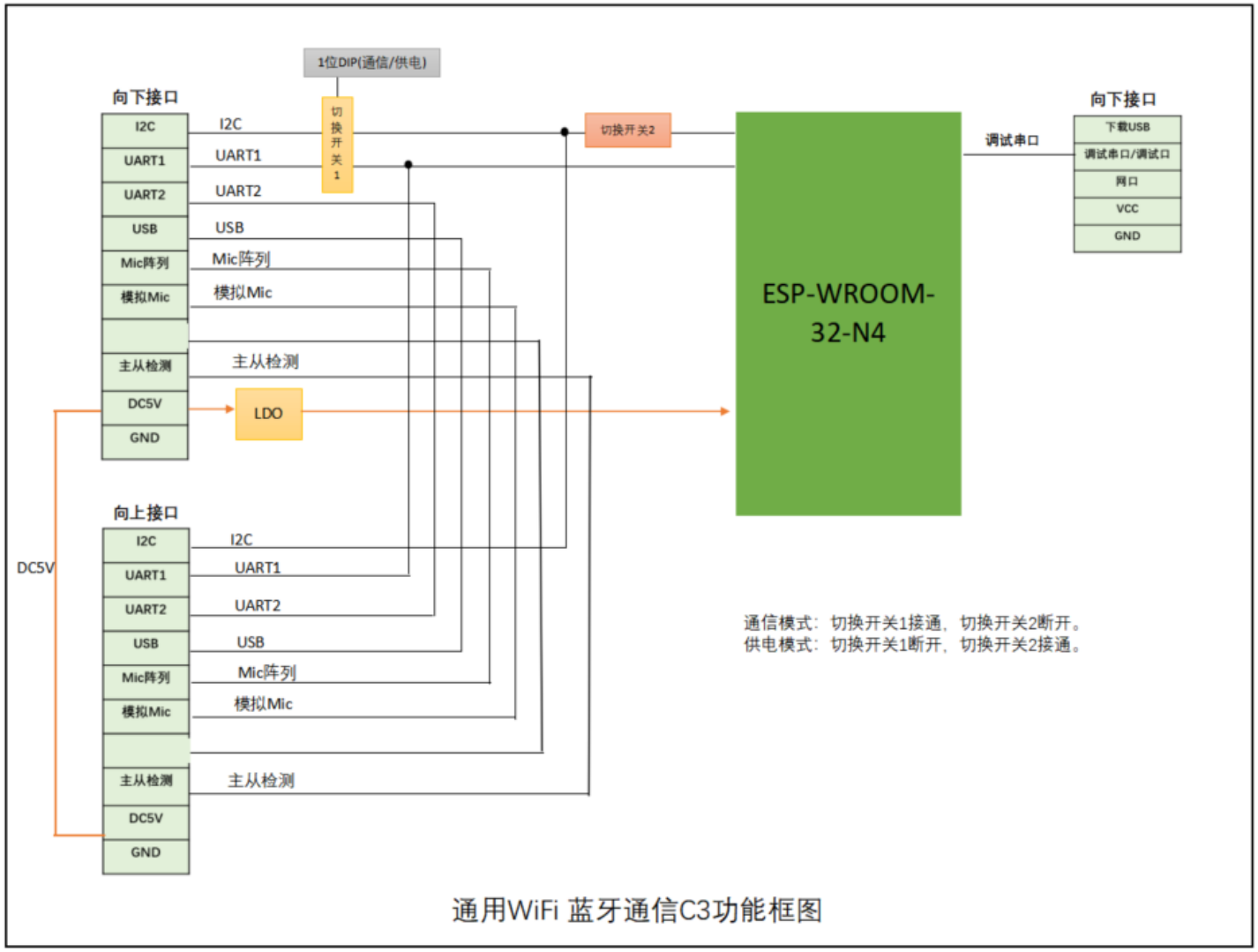




Zigbee通信C2功能框图

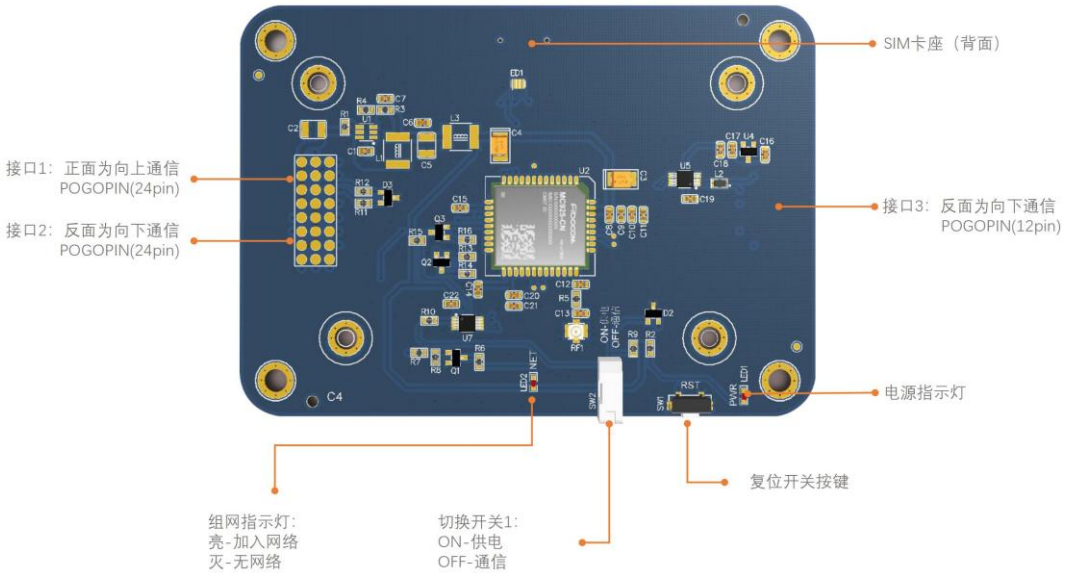
C3接口

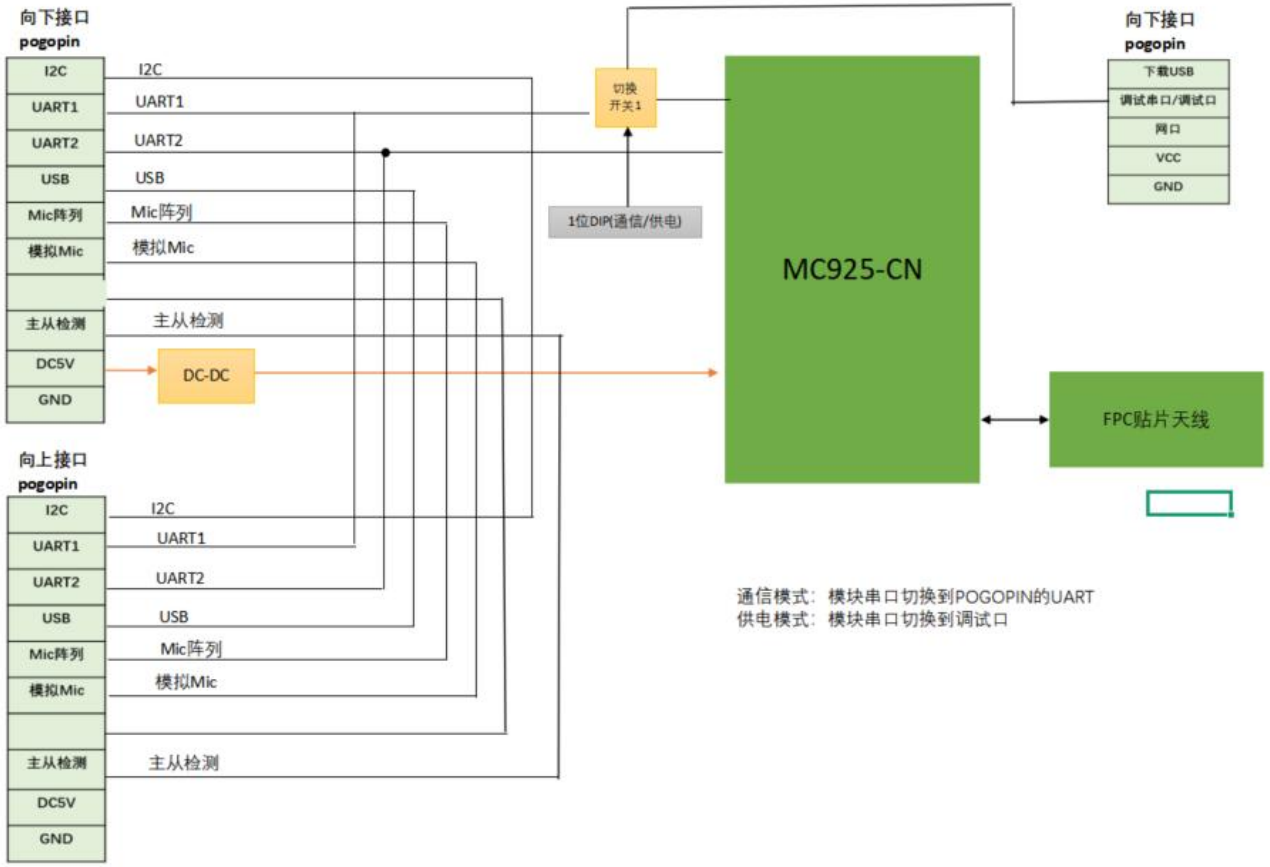




通用WiFi 蓝牙通信C3功能框图

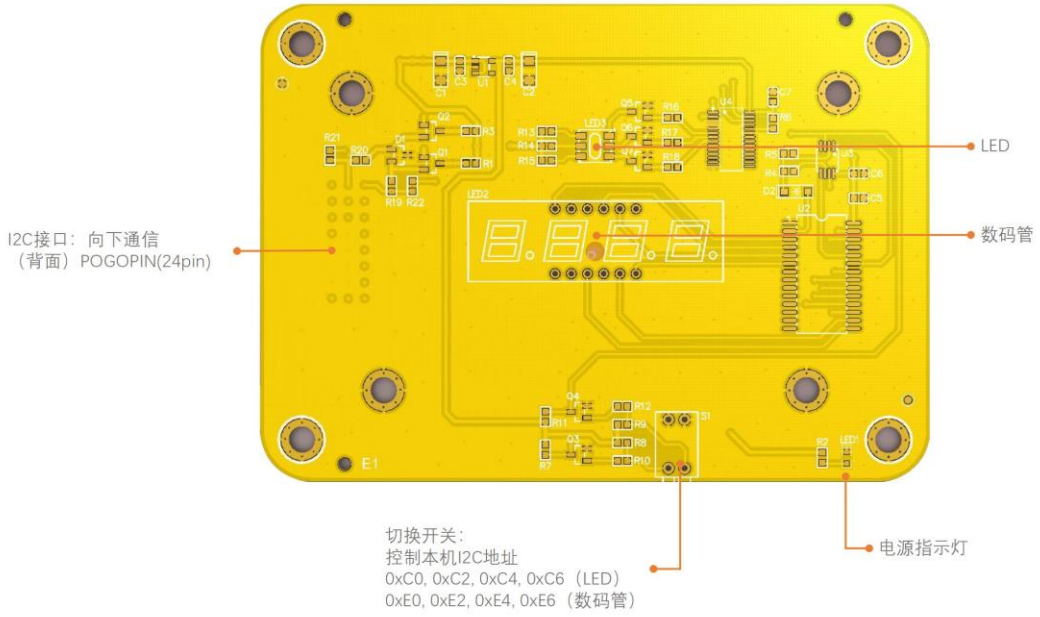
C4接口

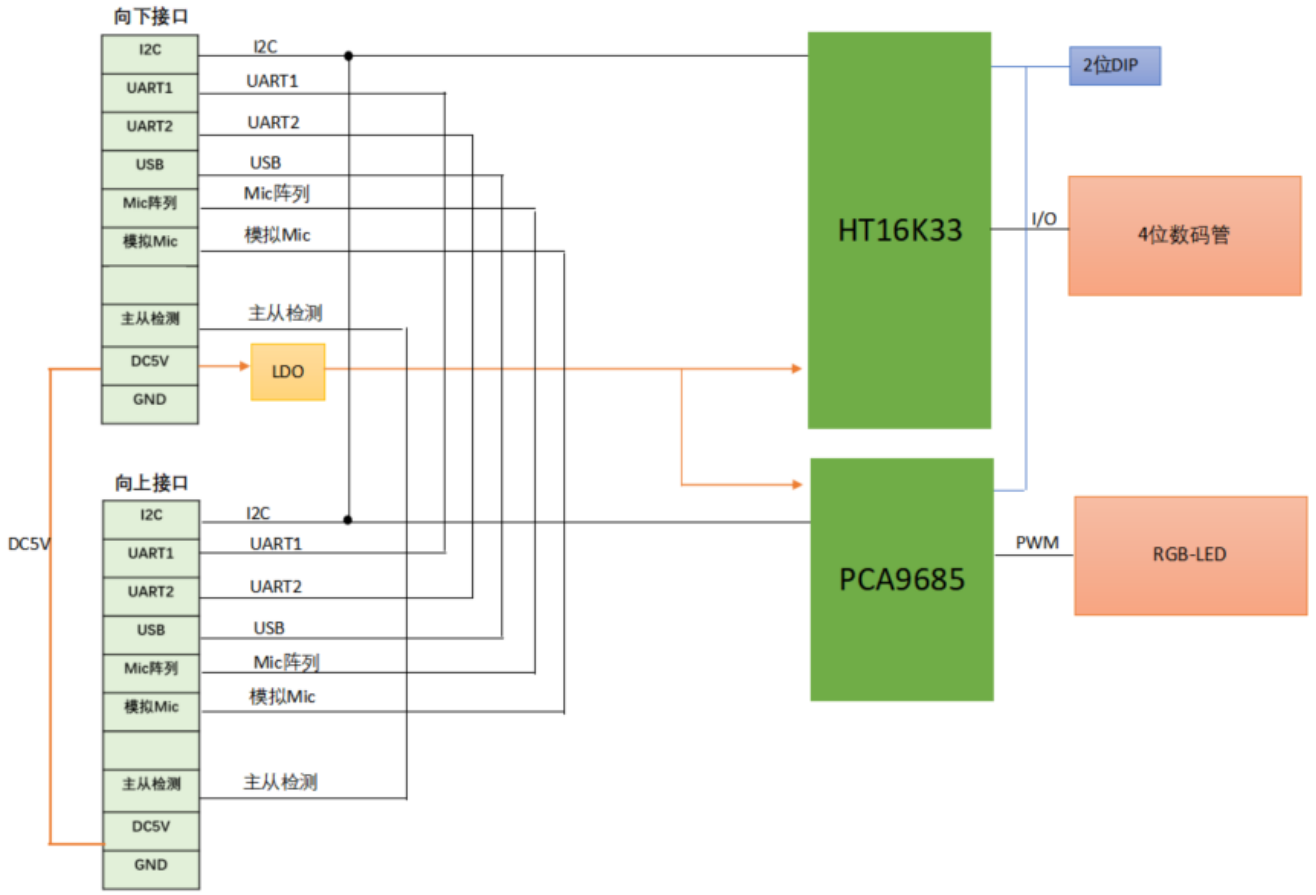




Nbiot通信C4子板功能框图

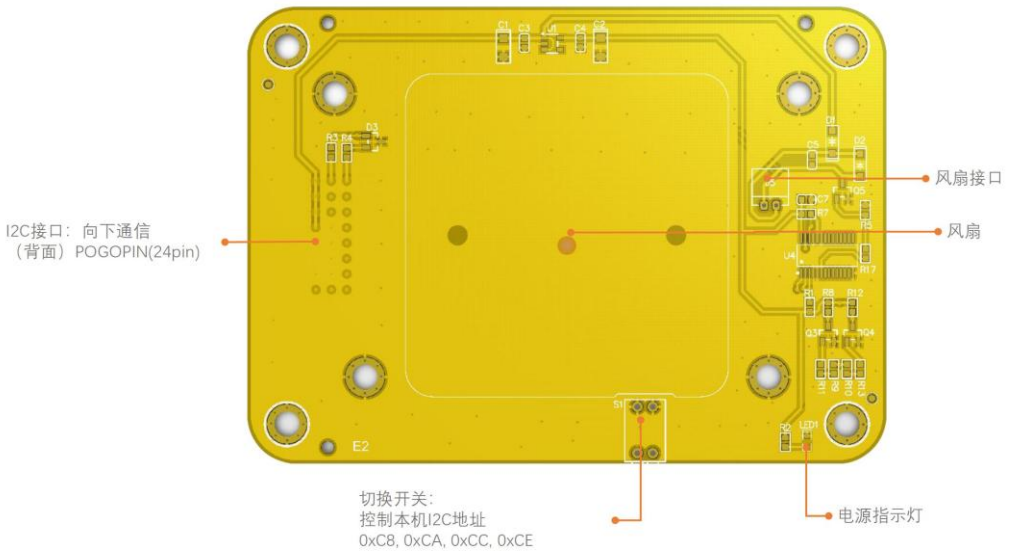
E1接口

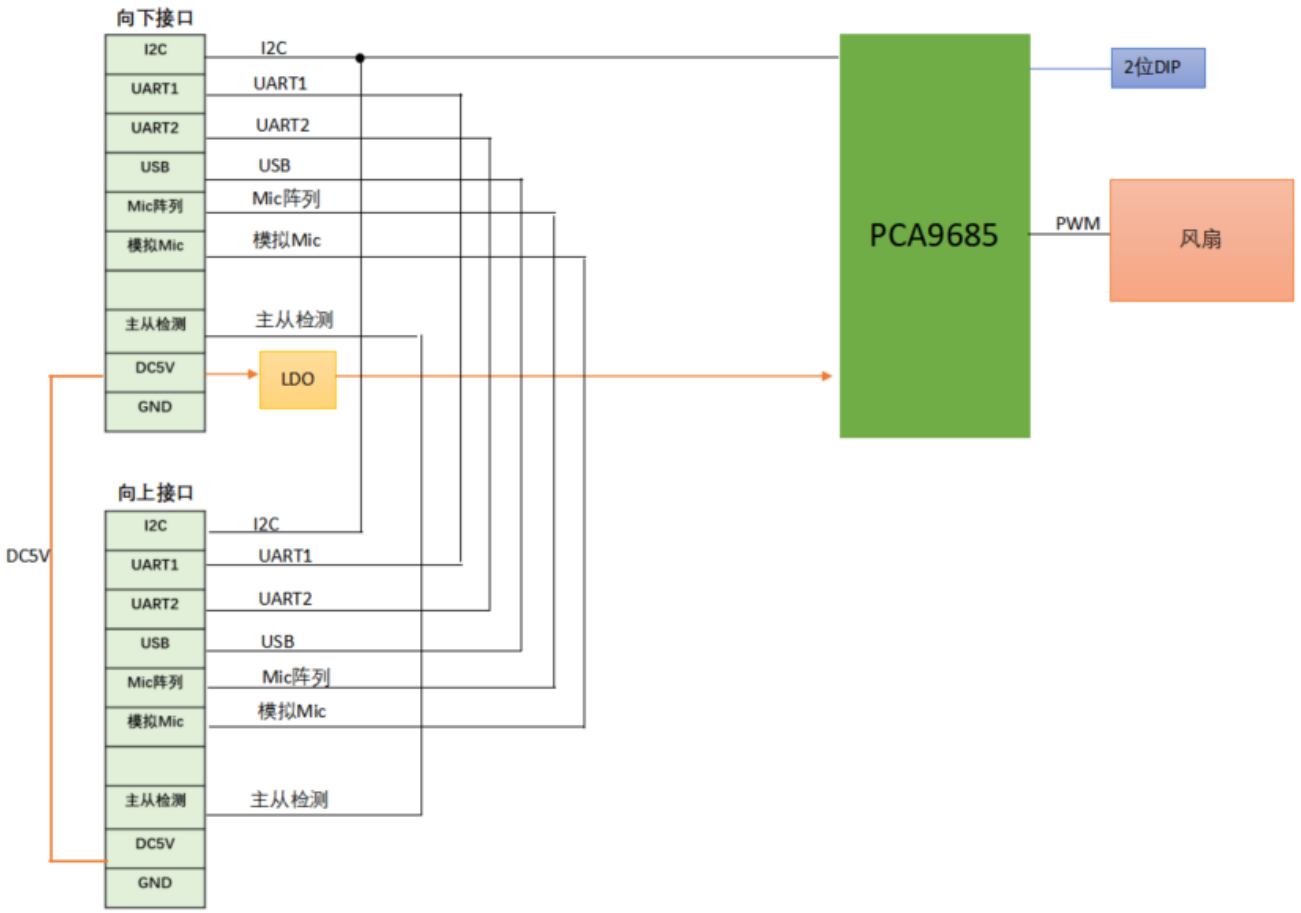




LED子板S1功能框图

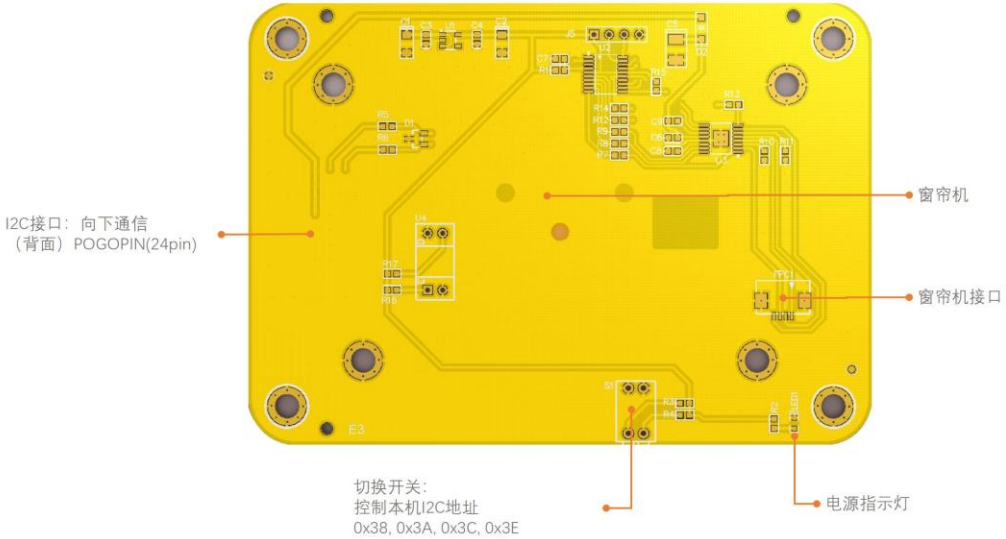
E2接口

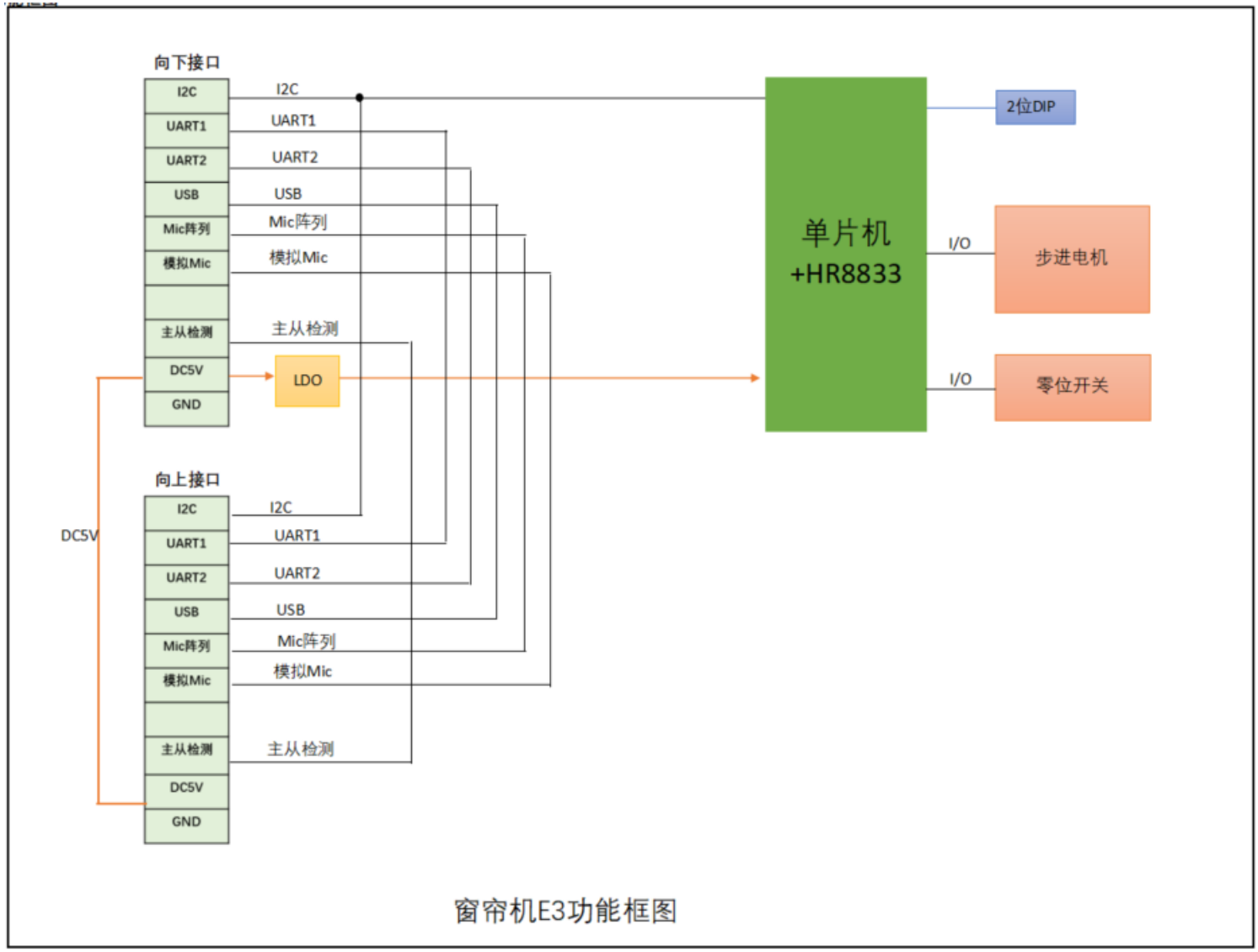




风扇E2功能框图

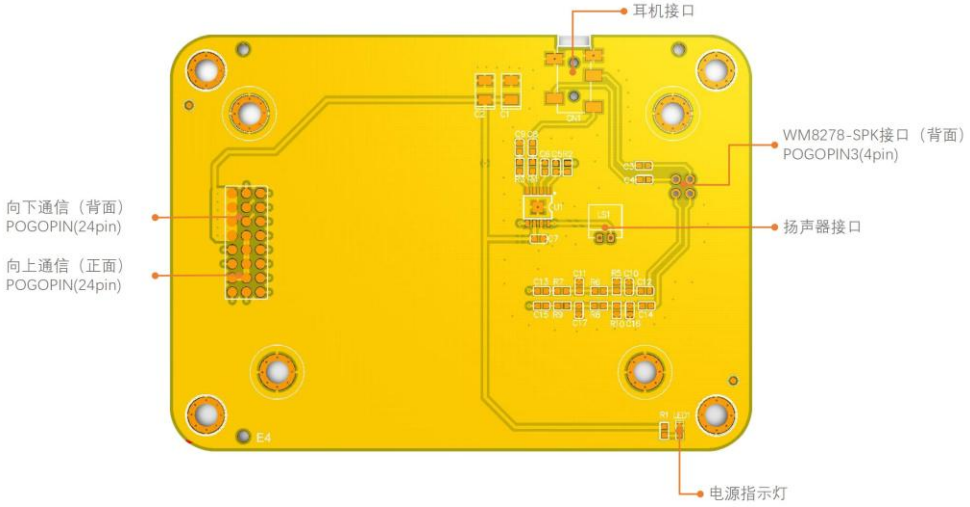
E3接口

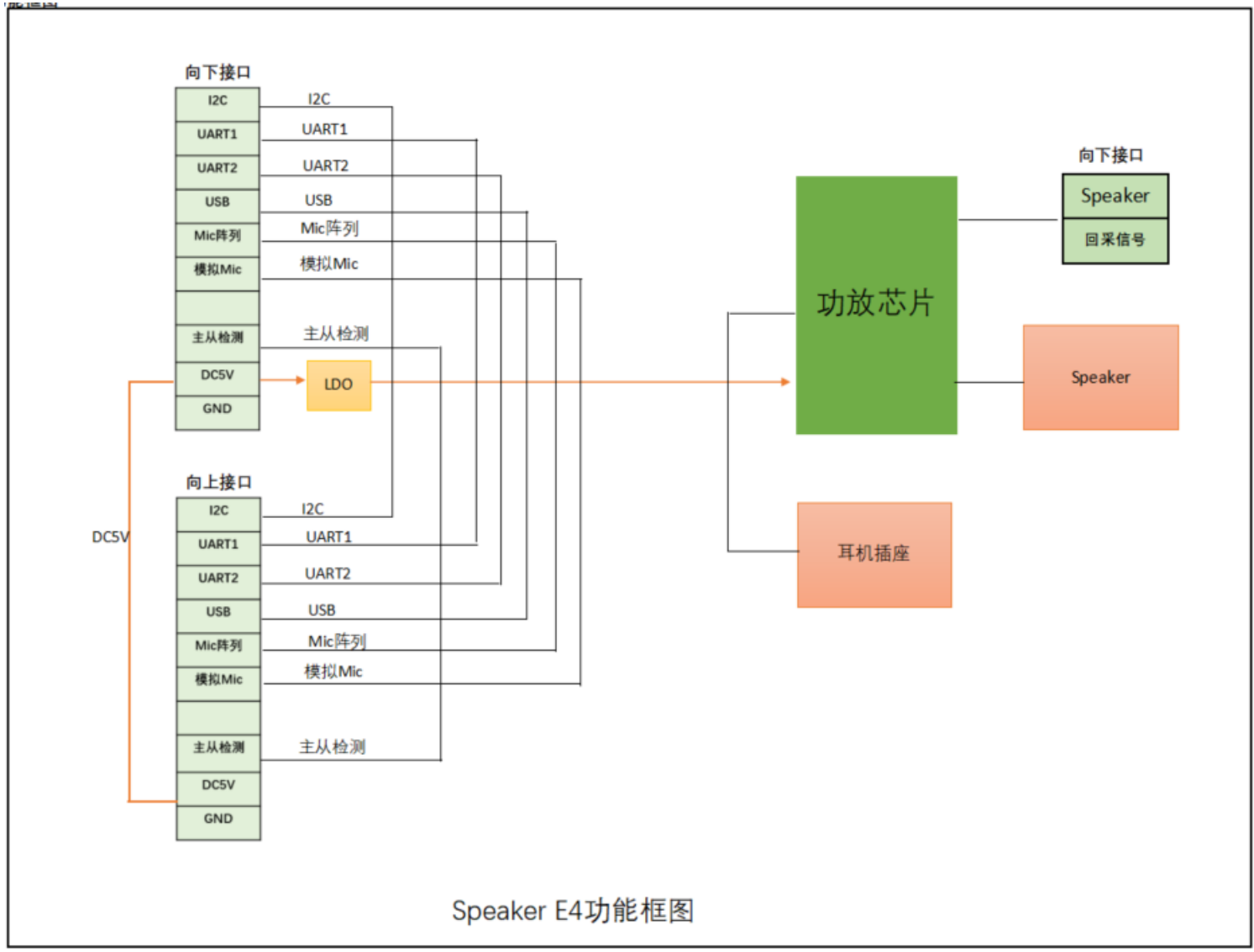




窗帘机E3功能框图

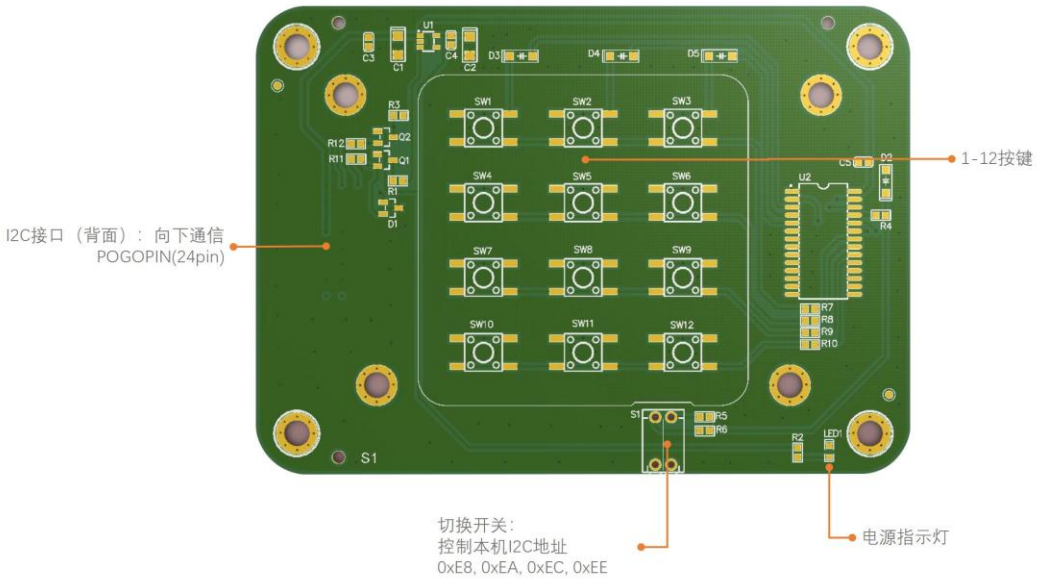
E4接口

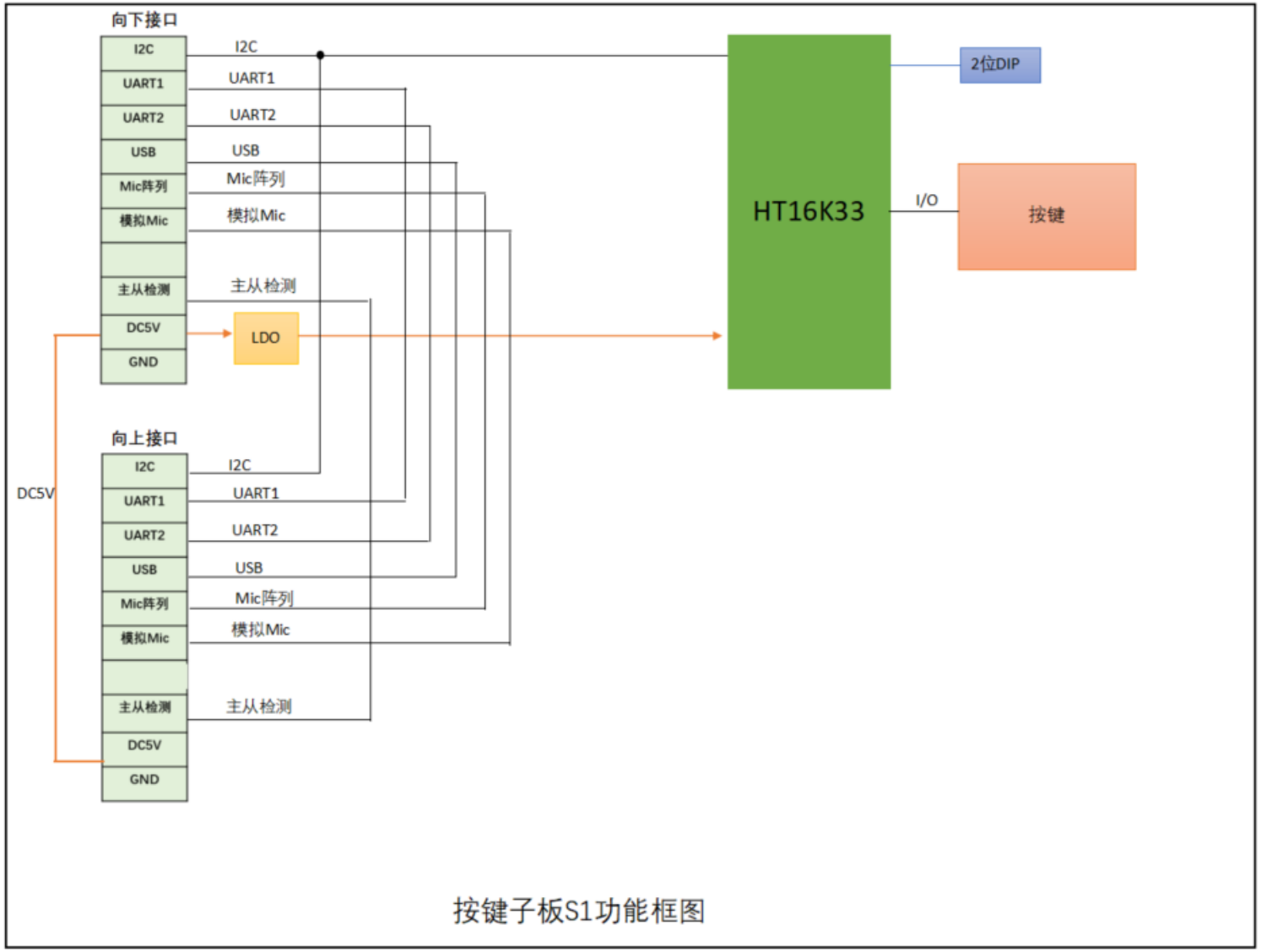




Speaker E4功能框图

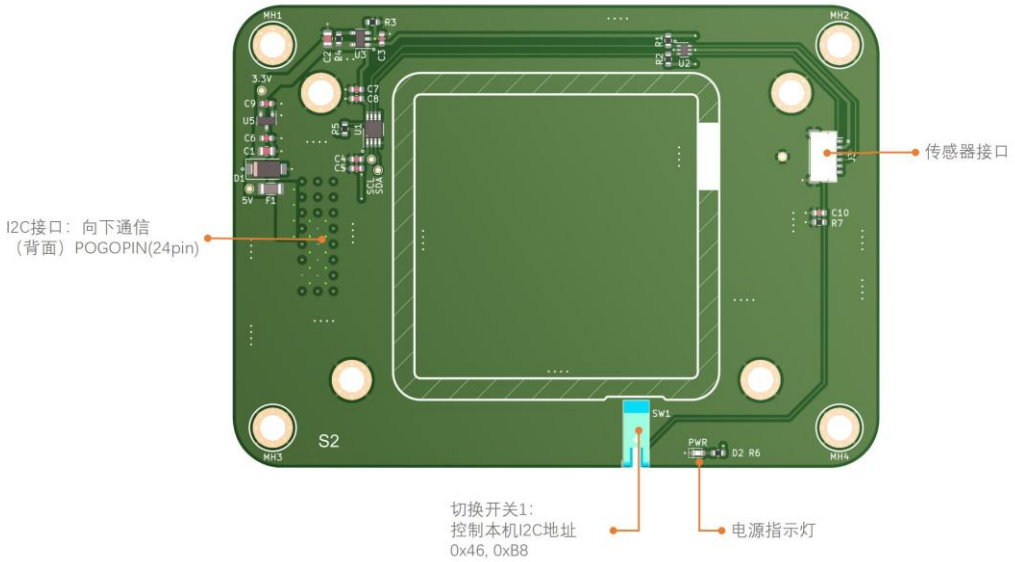
S1接口

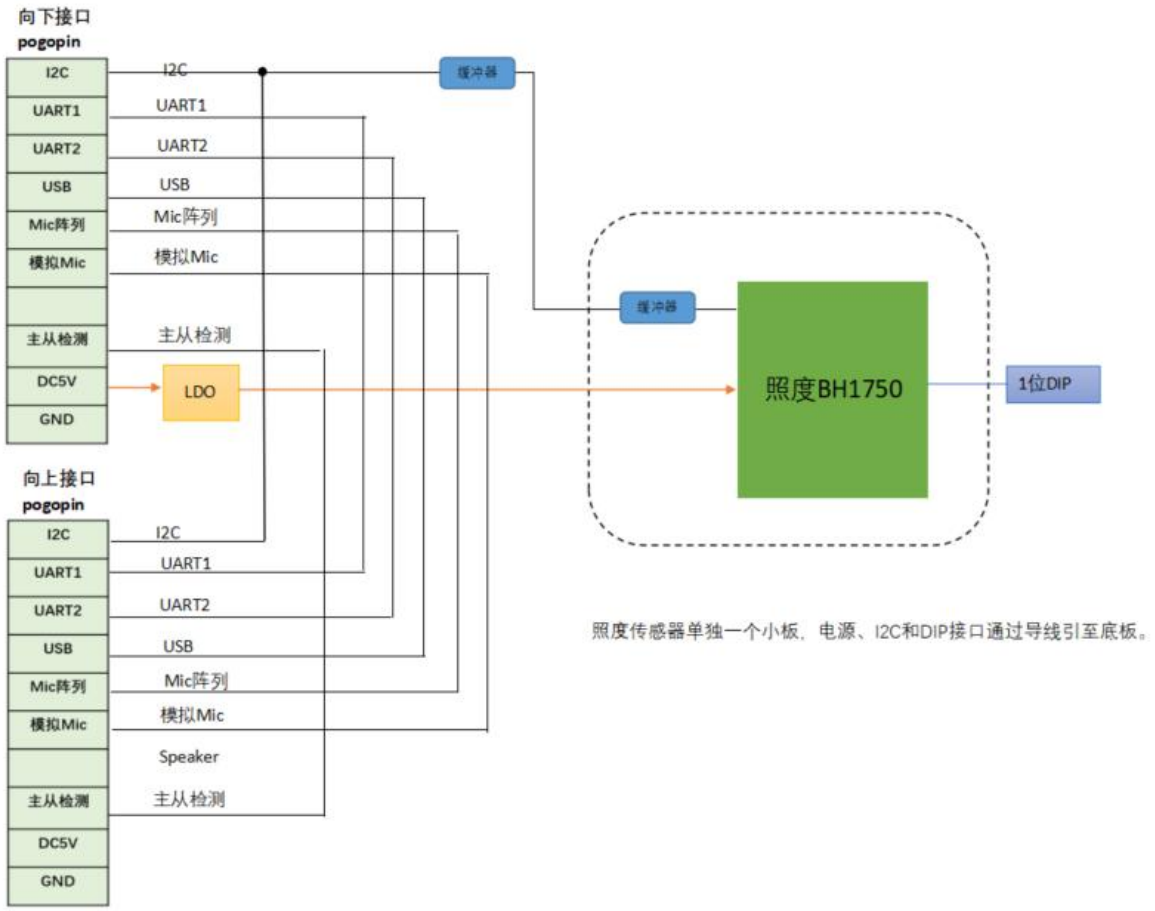




按键子板S1功能框图

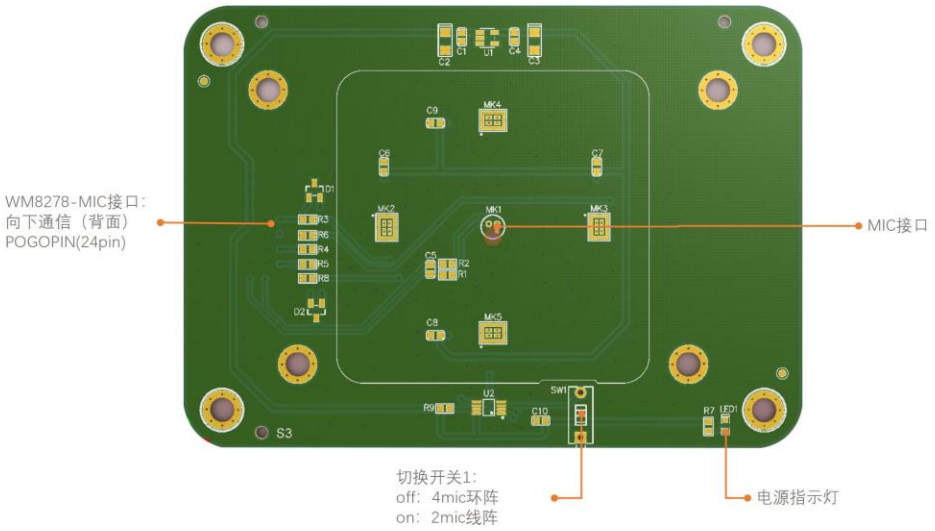
S2接口

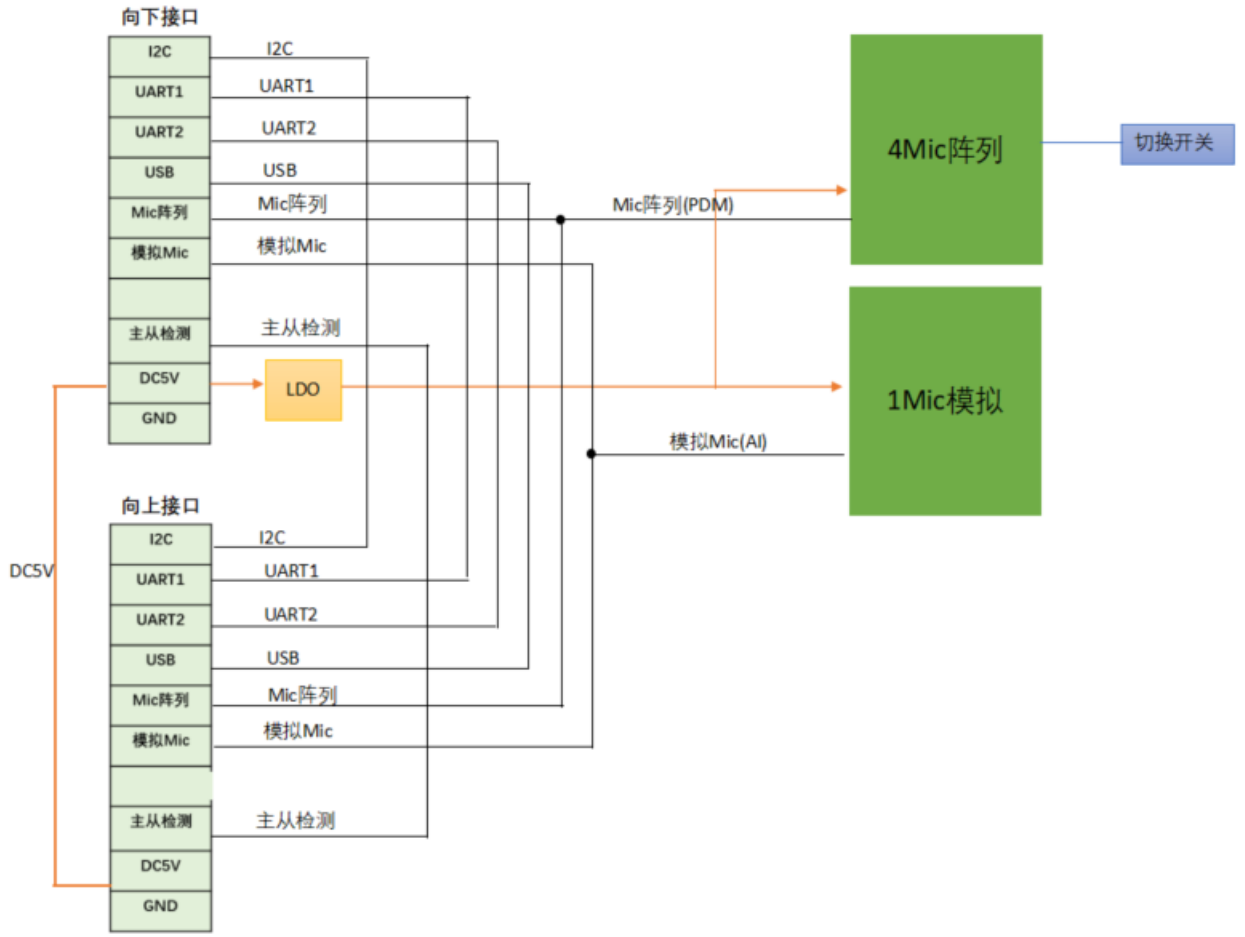




光照度传感器S2功能框图

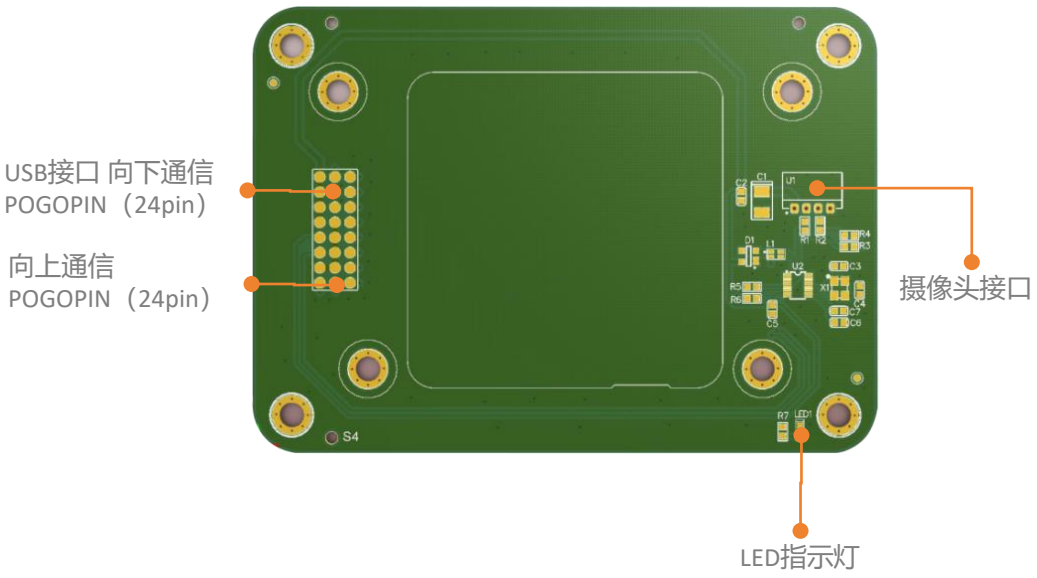
S3接口

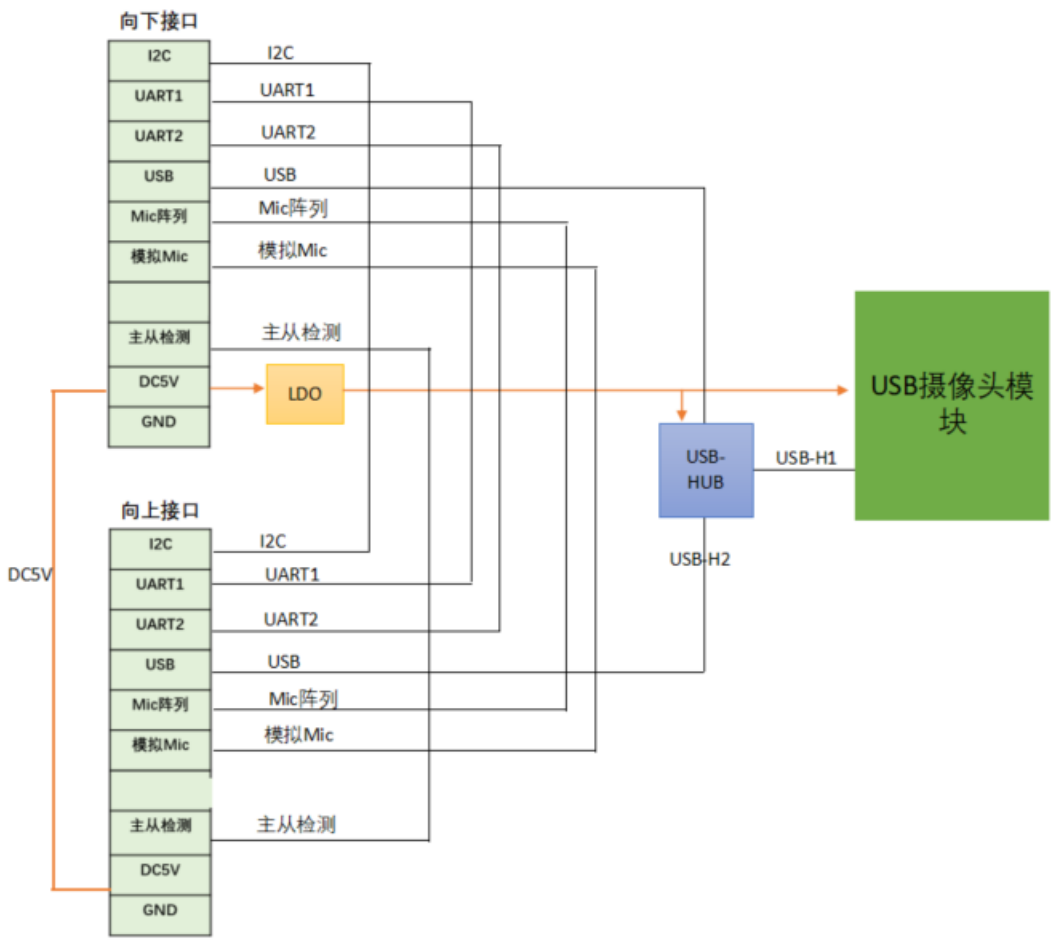




麦克风阵列S3功能框图

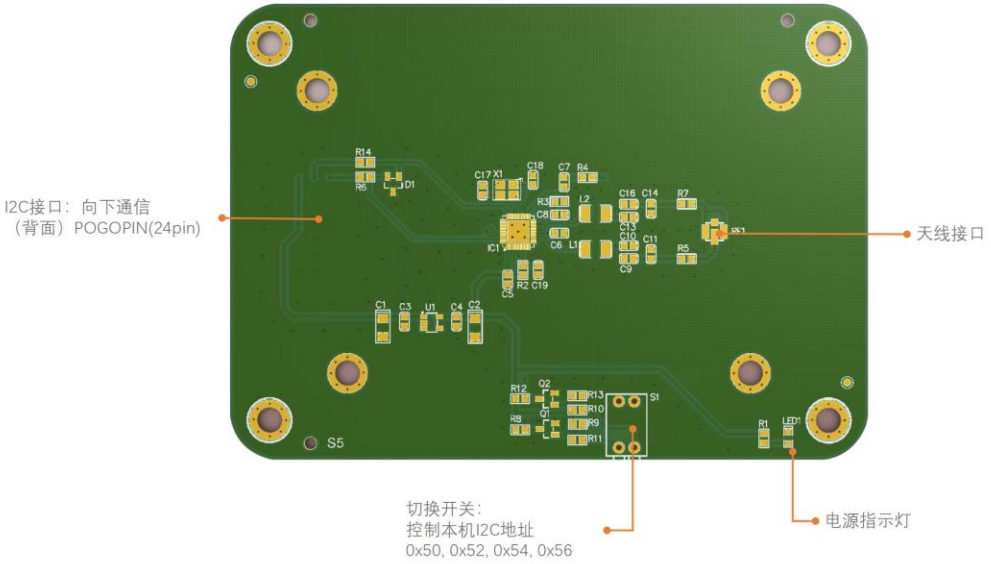
S4接口

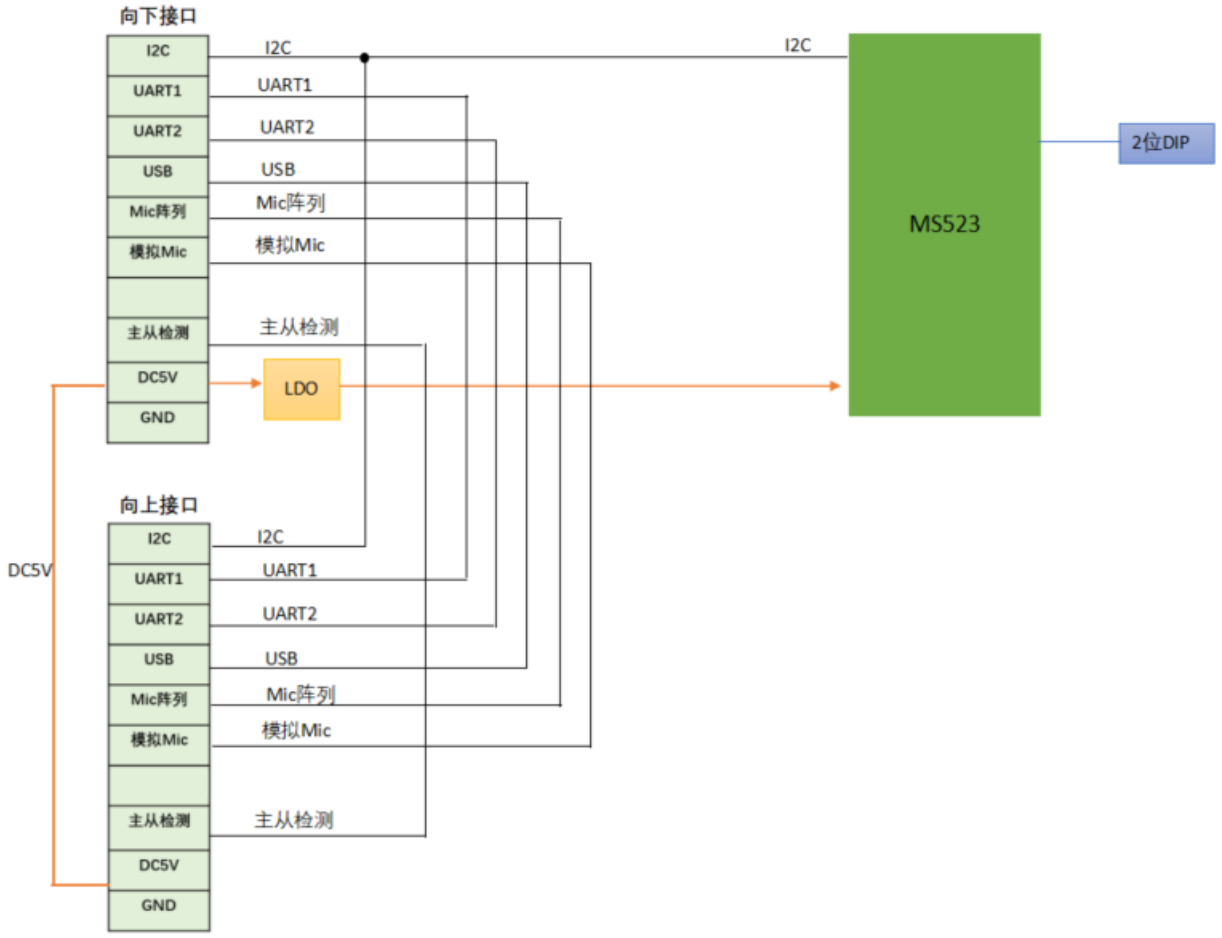




摄像头S4功能框图

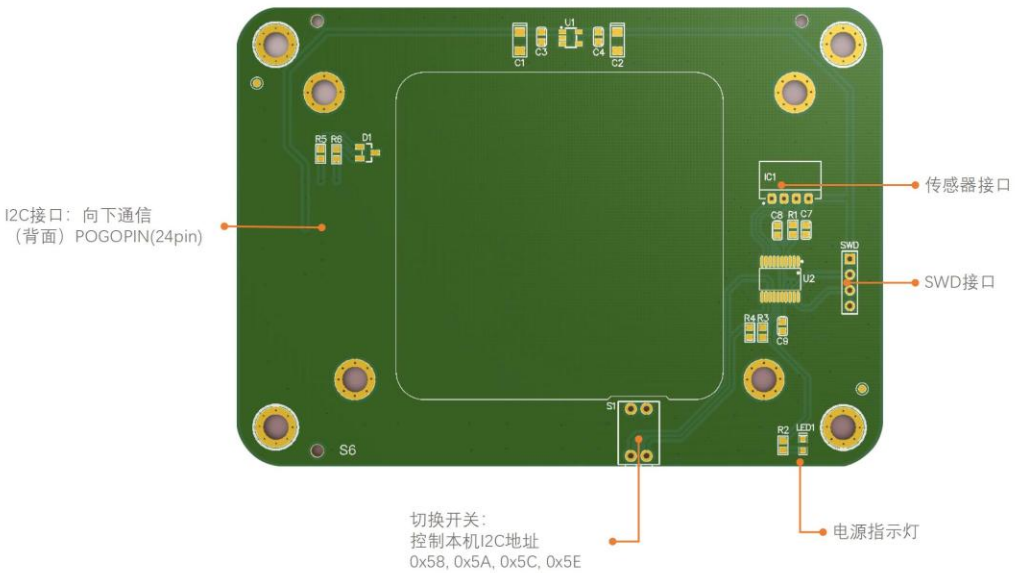
S5接口

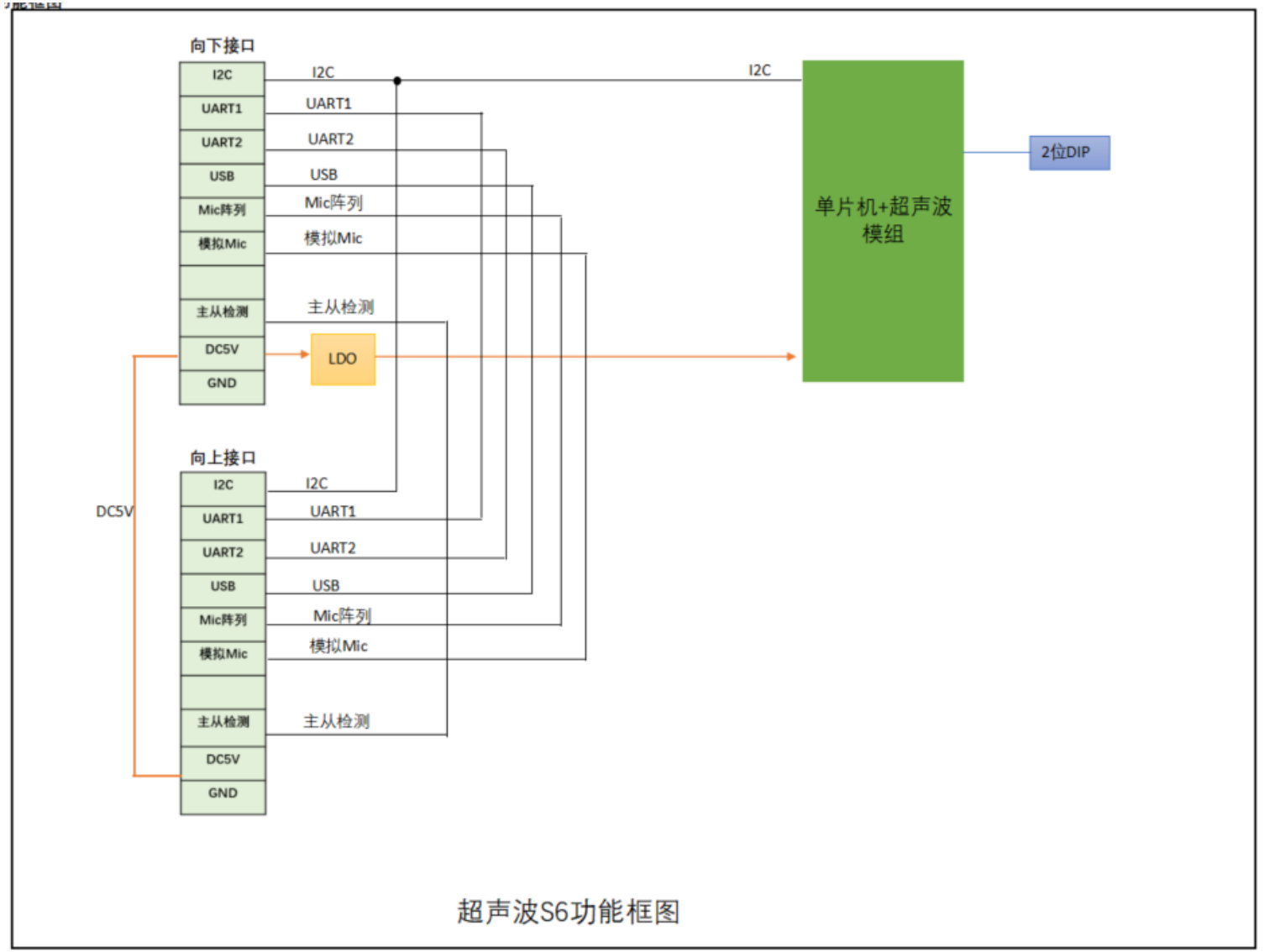




NFC子板S5功能框图

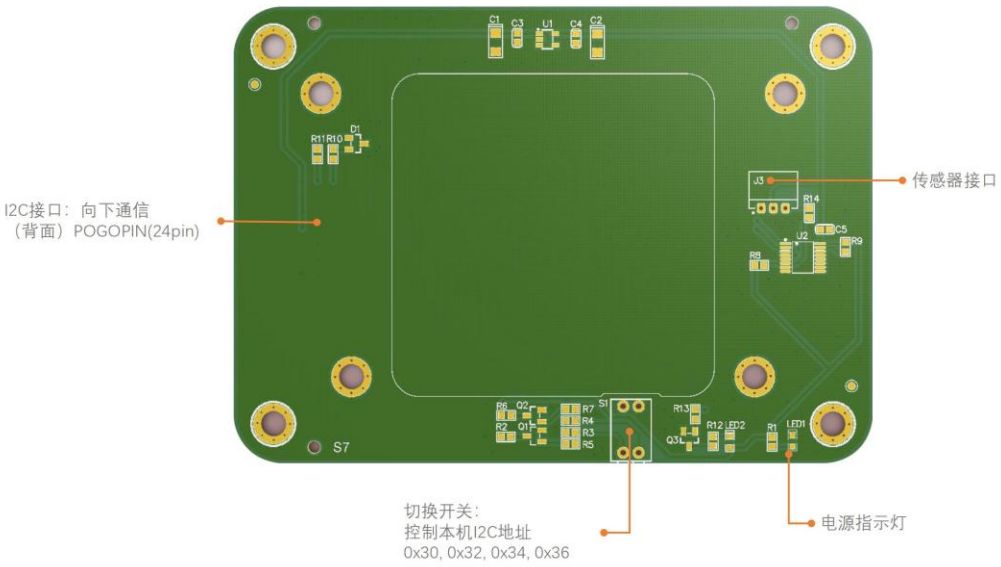
S6接口

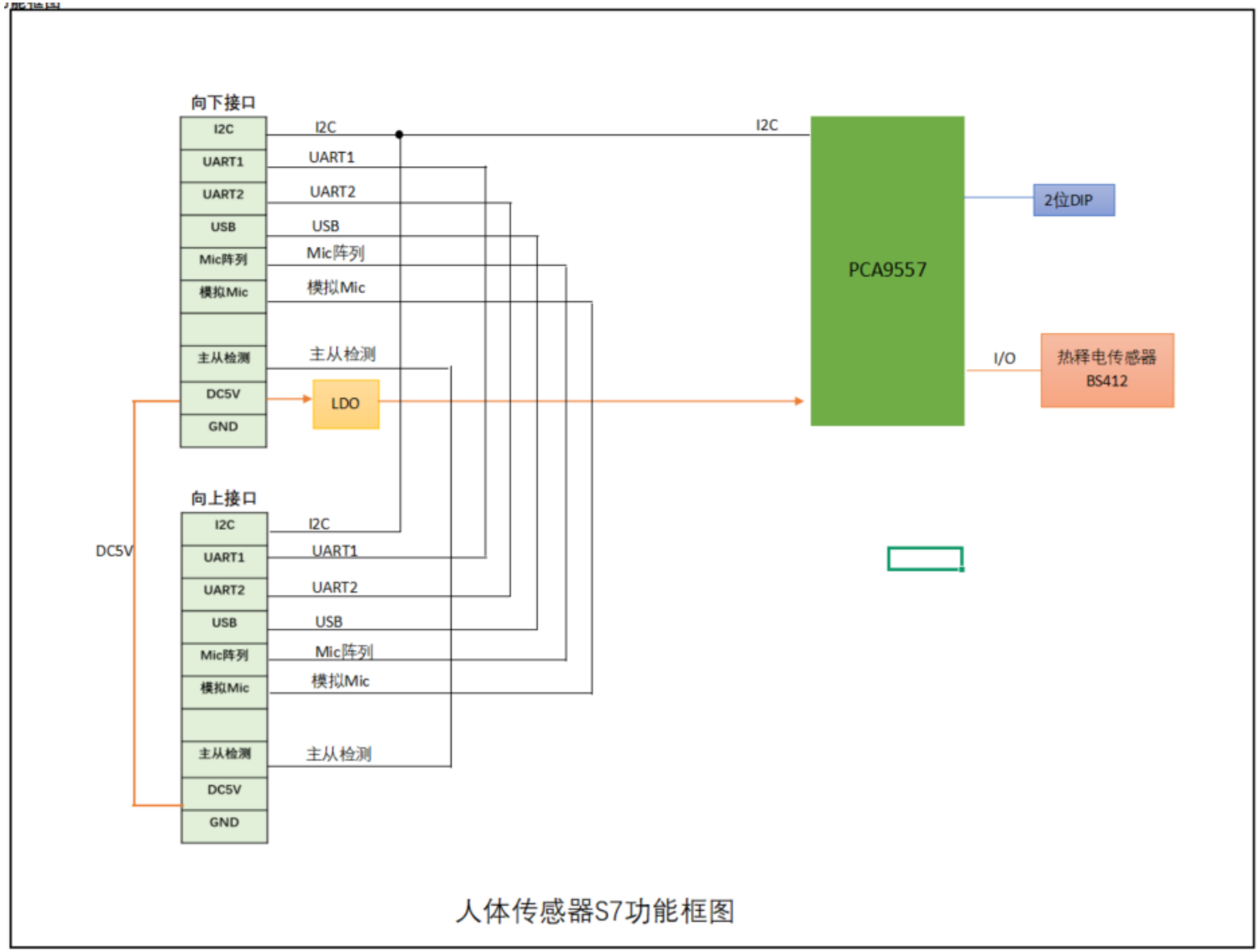




超声波S6功能框图

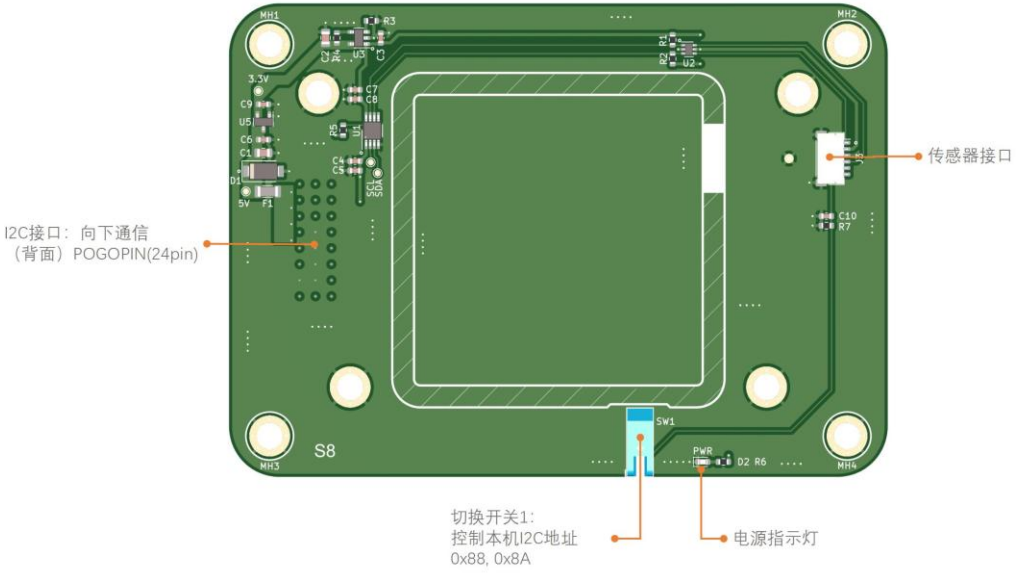
S7接口

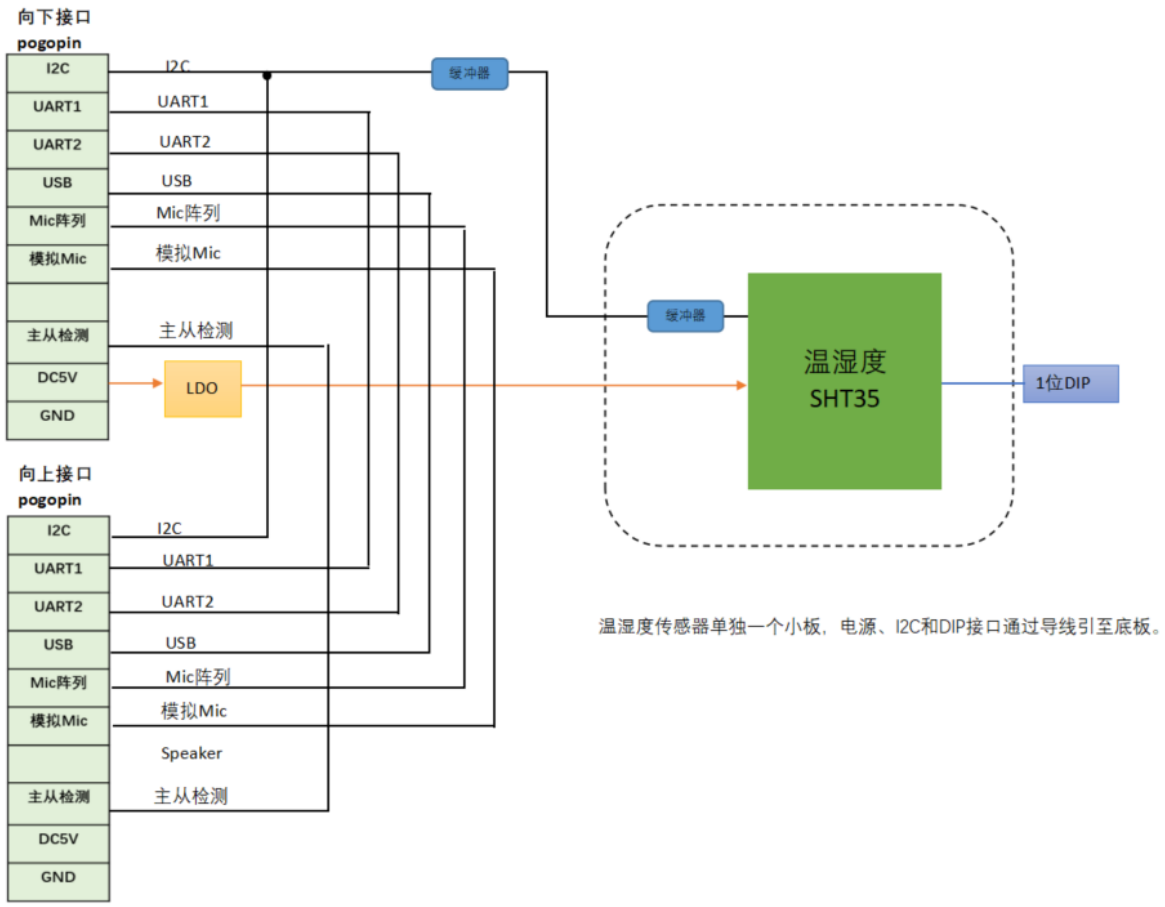




人体传感器S7功能框图

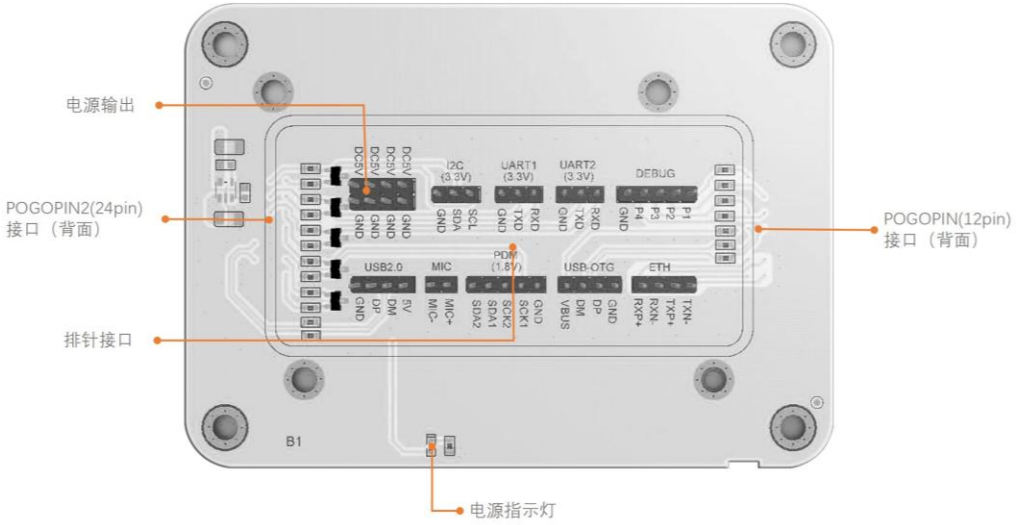
S8接口

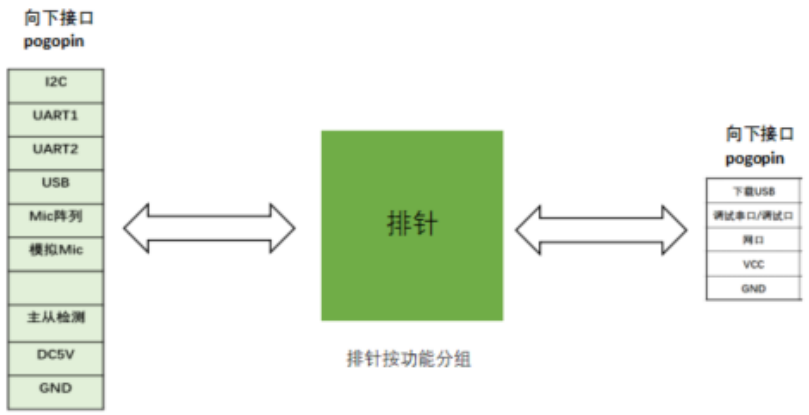




温湿度传感器S8功能框图

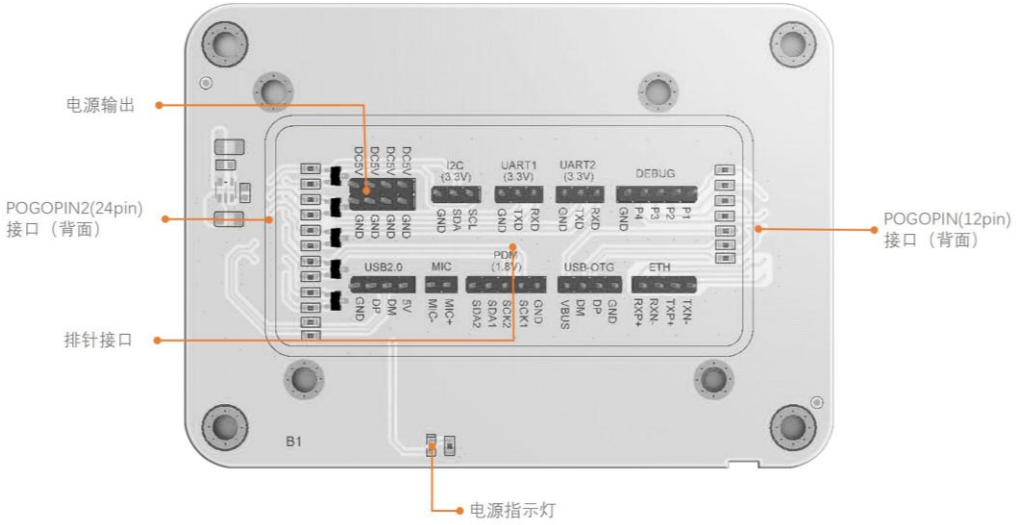
B1接口

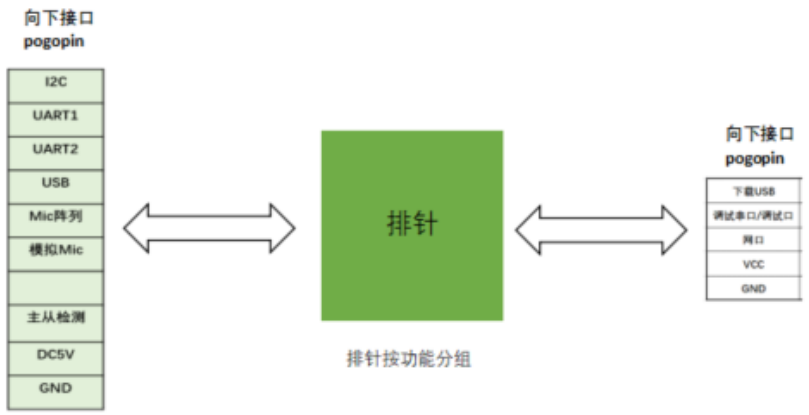




低速接口扩展B1功能框图

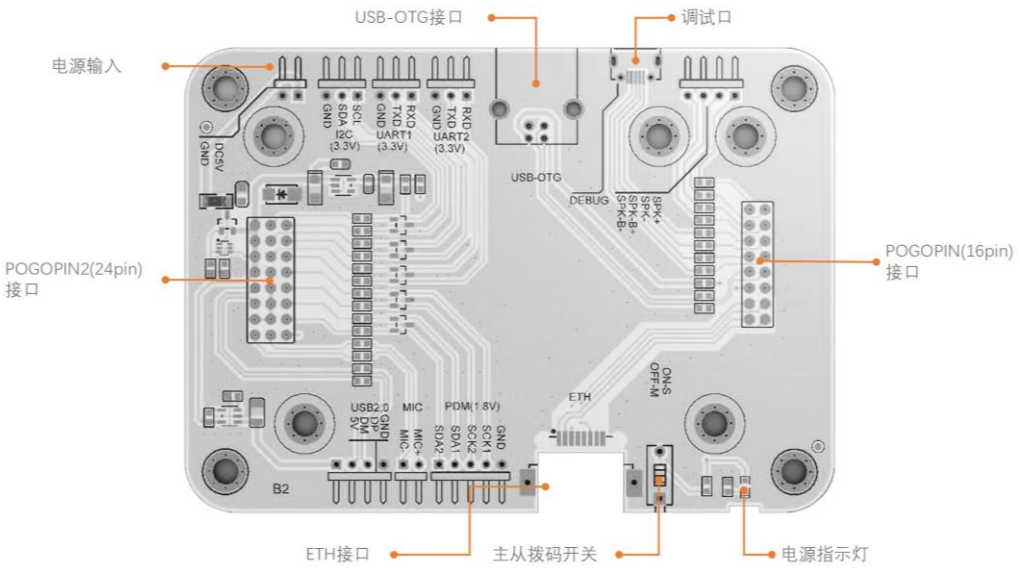
B1接口

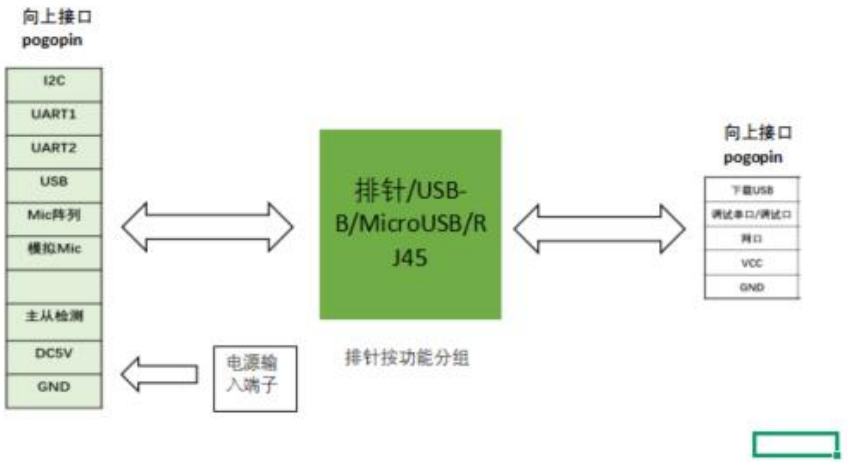




低速接口扩展B1功能框图

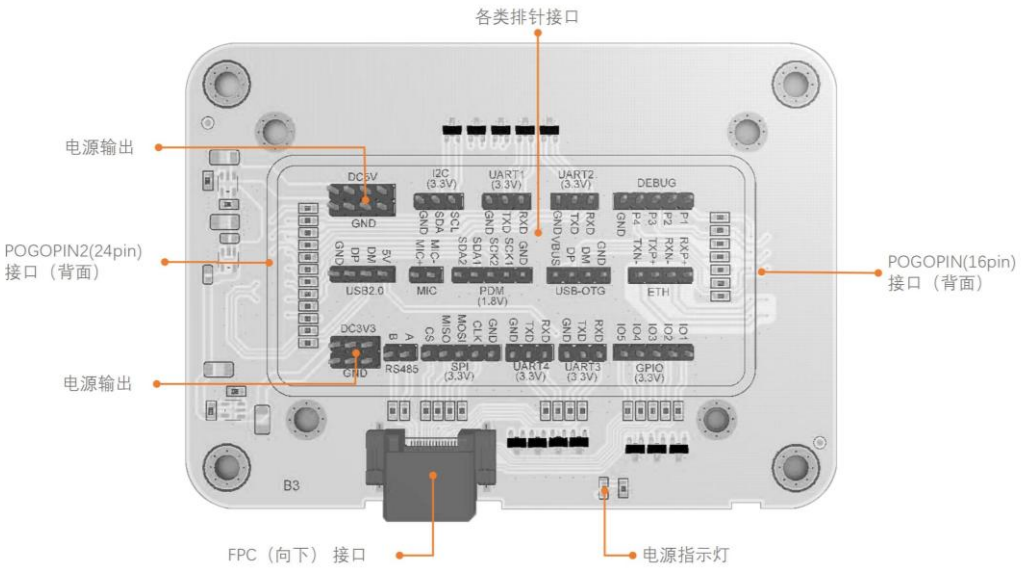
B2接口

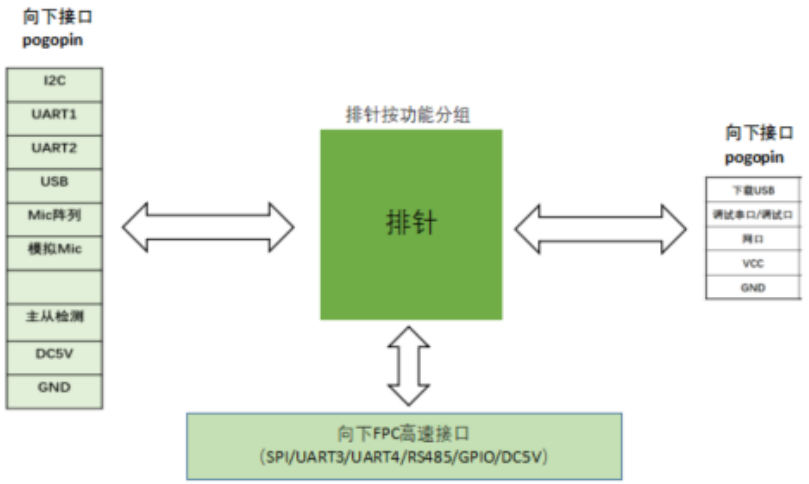




子板扩展B2功能框图

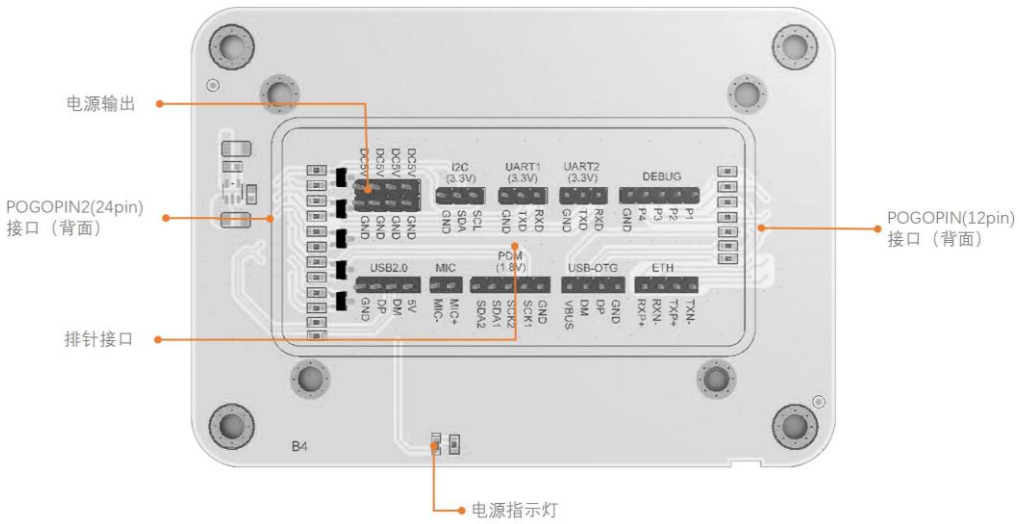
B3接口

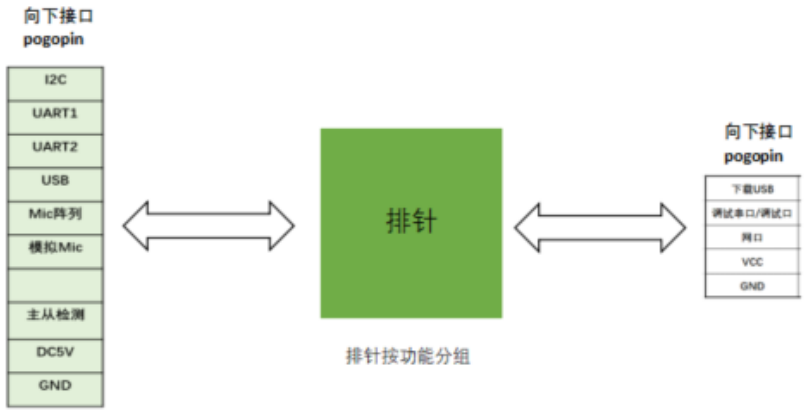




多协议接口扩展B3功能框图

B4接口





U板扩展B4功能框图